Some like it hot: Thermal Performance and Activity Patterns of Halictid Bees on Lesvos Island, Greece

Abstract

Approximately three-quarters of bee species are ground-nesting and spend much of their life cycle underground, where soil microclimate may be more important than other macro-environmental variables in shaping their ecology, behaviour, and life history. Despite this, the role of fine-scale thermal heterogeneity during early nest-founding stages remains poorly understood, particularly in primitively eusocial bees. Here, we combine physiological assays and behavioural observations to investigate how microclimatic variation influences thermal performance and activity patterns in foundresses of two ground-nesting halictid bees, *Halictus scabiosae* and *Lasioglossum malachurum*, during spring. First, following chill coma induction, we measured time to right, time to take-off, and body temperature during righting and take-off to assess species-specific differences in thermal thresholds and recovery capacity. Second, we quantified nest-associated behaviours and general activity at two nesting sites, while recording light, ambient temperature, and soil temperature across a depth gradient. Thermal assays revealed species-specific differences in performance, with *L. malachurum* exhibiting lower righting temperatures than *H. scabiosae*, although flight initiation traits were less differentiated. Activity at nesting aggregations was associated with air and soil temperature at all depths, showed non-linear responses across thermal gradients, and the relationships with shallow and middle soil temperature differed between sites. Abundance of *H. scabiosae* increased with shallow soil temperature and light intensity, whereas that of *L. malachurum* showed weak or no relationships. Nest guarding in *H. scabiosae* responded to subsurface temperature in opposite directions: guarding probability declined with shallow soil temperature, whereas guarding duration increased with soil temperature at 5 cm. Overall, our findings indicate that activity in ground-nesting halictid bees is associated with fine-scale microclimatic heterogeneity, and that the two focal species differ in righting temperature and in the sensitivity of their activity to microclimate. Understanding these dynamics under environmental change will require stratified soil measurements, quantification of biotic pressures, and individual-level tracking across sites, seasons, and colony phases.

Keywords: Halictidae, soil microclimate, early-season activity, chill coma recovery, Aegean islands

1 Introduction

Organisms adapt to changing environmental conditions not only through morphological and physiological changes at the individual level (González-Tokman et al., 2020; Rodrigues and Beldade, 2020), but also through adjustments in behaviour and energy allocation at the population level (Gould , 1978; Stollewerk et al., 2025). Environmental conditions can be both predictable and stochastic, and organisms must cope with this variability either by modifying their activity patterns and energetic investment in key fitness-related processes, such as growth, development, foraging, and reproduction (Eliopoulos and Stathas , 2003; Whitfield and Richards , 1992), or at the population level, by shifting their distribution (Davis and Shaw , 2001; Huang et al. , 2022; Lomolino et al. , 2017). Consequently, life-history traits shift across temporal and environmental gradients (Stearns , 1976). Rather than being finely tuned to a single set of conditions, organisms often exhibit phenotypic plasticity that allows them to persist in fluctuating environments (Burton et al. , 2022). Such environmental variability has given rise to a broad range of life-history strategies, shaped by natural selection and reflecting both contemporary ecological conditions and the evolutionary history of species. One such environmental factor that strongly influences organismal niches and life-history strategies is temperature (Gilbert and Raworth, 1996; Howe , 1967; Kamm , 1974; Melville and Schulte , 2001; Porter , 1988; Truebano et al., 2018). Thermal heterogeneity generates microclimatic variation within habitats, producing fine-scale temperature gradients that can constrain dispersal, influence gene flow, and potentially contribute to population divergence (Johansson et al. , 2016). It also influences physiological rates, thereby affecting performance, energy allocation, and the nature of biotic interactions, while driving morphological, physiological, and behavioural adaptations that enable organisms to cope with environmental variability (Abram et al. , 2017; Kingsolver and Huey , 1998; Tattersall et al. , 2012). Organisms have evolved ingenious strategies to cope with thermal variation so that key life stages occur when predictable thermal conditions are most favourable (Forrest and Miller-Rushing , 2010).

Ectotherms, characterised by low resting metabolic rates and high thermal conductance, are particularly vulnerable to environmental temperature fluctuations because they rely heavily on external conditions to regulate physiological processes and behaviours (Cossins , 2012). Consequently, a myriad of adaptations—genotypic and phenotypic, including morphological, physiological, and behavioural traits (Bale et al., 2002; Dunham , 1993; Helmuth et al., 2005)—have evolved to cope with temperature variability, whether predictable (e.g., diurnal or seasonal cycles) or stochastic (e.g., extreme weather events). Ground-nesting ectotherms, such as ants, wasps, and bees, occupy the interface between heliothermy (heat gain from solar radiation, e.g. basking in sunlight) and thigmothermy (heat exchange with the ground or nest substrate, largely through conduction), making them particularly interesting systems for examining the thermoregulatory strategies they adopt to cope with thermal fluctuations under different environmental conditions.

The relationship between thermoregulation and social organisation can be further complicated because social strategies may be obligate or facultative depending on environmental conditions (Richards et al., 2003; Soucy, 2002). Additionally, multiple broods within a season may exhibit differing strategies depending on their emergence time, highlighting the complexity of behavioural adaptations. Behavioural thermoregulation, through adjustments in activity patterns and the

frequency, duration, and timing of behaviours such as nest closing and nest guarding in response to variation in temperature and solar insolation across microclimatic mosaics, emerges as a necessary buffering strategy (Bartholomew , 1964; Bogert , 1949; Kearney et al., 2009), especially in ectotherms that operate close to their thermal limits (Huey and Stevenson , 1979; Kearney et al., 2009).

The sweat bee family Halictidae (Hymenoptera), in which social organisation varies within and among species, therefore offers a tractable system for examining how behavioural strategies are shaped by thermal constraints. This large and taxonomically diverse group includes many ground-nesting bees. More broadly, approximately three-quarters of wild bee species nest in the ground (Antoine and Forrest , 2021; Fawcett et al., 2019; Knerer , 1969). Halictidae exhibit remarkable social polymorphism, spanning a continuum from solitary and communal to semi-social, primitively eusocial, and advanced eusocial forms (Dunn and Richards , 2003; Eickwort et al., 1996; Richards , 2000), with the limits of each category remaining unclearly defined (Antoine and Forrest , 2021). Halictid bees are present across most faunal zones and consist of closely related species, suggesting ongoing evolutionary diversification within the group (Richards , 1953). However, even closely related species may differ substantially in nest architecture (Knerer , 1969) and, subsequently, in social organisation (Packer and Knerer , 1985; Schwarz et al., 2007), and these species can often be found nesting side by side in the same location (Batra , 1966; Knerer , 1969). Colony size ranges from small (2–4 individuals) to large (over 200 individuals), sometimes with strong division of labour and caste segregation (Lienhard et al., 2010). In some species, multiple foundresses may cooperatively establish a single nest (pleometrosis), after which one female becomes the dominant reproductive individual (Packer , 1986). Subordinate females may remain as workers or disperse to establish new nests as colony size increases. Nest reuse across seasons is common, with females often overwintering near their natal nests (Kukuk , 2002). The diversity in social organisation makes halictid bees a valuable system for examining how sociality interacts with thermoregulatory behaviour. In particular, variation across generations and nesting environments, such as ground-nesting conditions influenced by soil temperature and structure, provides an opportunity to investigate how behavioural and physiological responses to temperature are shaped by both social and environmental factors.

Early-season pollinator activity remains understudied, particularly in ground-nesting bees and in Mediterranean settings (Harmon-Threatt , 2020). Bees are among the most important pollinators globally, and polylectic halictid bees (i.e. generalist foragers that do not specialise on specific host plants but utilise a wide range of floral resources) visit diverse wild and cultivated plants and play a crucial role in maintaining ecosystem functioning (Corbet et al., 1993). Evidence suggests that prior heat exposure does not substantially improve bees' ability to withstand high temperatures (Gonzalez et al., 2024). If spring-active reproductive foundress females are adversely affected by thermal stress, subsequent generations may fail to develop, potentially reducing population size and compromising long-term reproductive success and viability. Behavioural thermoregulation may therefore represent a key mechanism by which bees buffer temperature variation, making it essential to quantify how individuals allocate time to different activities under variable thermal conditions in natural settings.

This study focuses on two primitively eusocial, ground-nesting halictid bee species, *Halictus*

scabiosae and *Lasioglossum malachurum* (Michener , 2007; Plateaux-Quénu , 2008), which co-occur on the Greek island of Lesvos. Both species produce more than one brood per year (Fawcett et al., 2019; Michener , 2007), and are polylectic (Bogusch et al., 2020; Shebl , 2012). Because spring-active foundresses establish nests and produce broods, their ability to recover from cold exposure may be relevant to their early-season activity and nesting behaviours. We address a gap in early-season knowledge by quantifying the thermal performance and activity patterns of spring-active foundresses, which may differ from those of later broods that have been the primary focus of previous research (Gonzalez et al., 2024). By combining behavioural observations with experimentally derived thermal performance metrics, this study links experimentally measured recovery traits with realised behavioural responses in natural environments. Field observations (at a broad spatial scale) and nesting behaviours (at a focal-view scale), including nest entry and exit frequencies and nest-guarding duration, are combined with physiological performance measured using a chill-coma recovery assay. This assay records the time and temperature required for bees to regain righting ability and flight initiation (take-off), providing two measures of functional recovery from chill-coma impairment. We ask whether variation in recovery performance is reflected in activity and nesting behaviour under variable microclimatic conditions, and whether activity is better associated with air temperature, light, or soil temperature at different depths. We hypothesise that the less cold-resilient species maintains higher activity during warmer periods of the day and at higher ambient and soil temperatures. We also ask whether differences in body size may help explain variation in thermal performance between the two species. More broadly, these comparisons may help clarify how species-specific responses to temperature could shape activity during spring as temperatures change.

2 Methods

2.1 Synopsis of bee species

L. malachurum (Kirby, 1802) is a primitively eusocial halictid bee and is widely distributed in the Western Palearctic region across Europe and North Africa and from Iberia to the Caucasus (Knerer , 1992; Michener , 1979; Polidori et al., 2010). *L. malachurum* is smaller than *H. scabiosae* (Rossi, 1790) which is present in the west of the Palearctic and has its primary distribution in western-southern Europe but has recently been recorded in eastern-central Europe (Gil-Tapetado et al. , 2024; Pauly et al. , 2016). Both *L. malachurum* and *H. scabiosae* are primitively eusocial, and social organisation within Halictidae ranges from solitary to eusocial depending on ecological conditions and population structure (Gonzalez et al., 2024a). Both focal species are characterised by long flight seasons from early spring (March) till the start of autumn (September-October) (Brand and Chapuisat , 2012; Gogala , 1991; Knerer , 1980; Weissel et al., 2006). Primitively eusocial species of sweat bees like the two focal species are generally characterised by small colonies, ill-defined castes, frequent queen displacements and a higher proportion of males in the summer brood (Knerer , 1969). *L. malachurum* shows a relatively strict eusocial structure, with a clear reproductive division of labour and temporal separation between brood generations. However, the number of broods produced each season is more flexible, showing clinal variation and tending to increase with higher soil temperatures, although this effect is relatively weak and

is also associated with shorter nesting cycles (Weissel et al., 2006). Mated female gynes hibernate during winter before founding nests solitarily in spring. The number of broods laid can vary, with populations in warmer climates generally producing more broods per season (Knerer , 1992; Richards , 2000). In contrast, *H. scabiosae* exhibits a more flexible form of eusociality, with two distinct broods each season. Gynes overwinter in the same nest and, upon emerging from hibernation, reuse it for the following season (Lienhard et al., 2010). At the start of the season, females jointly initiate nests (Gogala , 1991), after which dominance hierarchies form and a single female typically becomes the primary reproductive individual while the others act as subordinates. Early-season broods are predominantly female (Knerer , 1967) and contribute to colony growth, whereas late-season broods consist of both males and gynes. Males die at the end of the season, while gynes mate in autumn, overwinter and become nest foundresses in the following year (Gogala , 1991).

2.2 Study site

The study was conducted in an unmanaged pasture near Skala Kallonis (39°12'36" N, 26°12'12" E; 1.0 m a.s.l.) on the Greek island of Lesvos over 10 days (15–24 April 2026), corresponding to spring conditions in the region (Figure 1, Supplementary Table S1, Supplementary Table S2, Supplementary Table S3). Previous work at the same study site has also examined both species in relation to thermal sensitivity (Gonzalez et al., 2020), although during summer conditions. The site consists of bare, sandy alluvial soil with sparse vegetation, providing suitable nesting habitat for ground-nesting bees (Gregory and Wright , 2005; Potts and Willmer, 1997; Severns , 2004; Wesserling and Tscharntke , 1995; Wuellner , 1999). The area also includes patchy anthropogenic disturbance, such as metal fencing, wire, and wooden debris, consistent with previous observations that halictine bees (members of the subfamily Halictinae within the family Halictidae, including the two focal species examined in this study) may preferentially nest in disturbed habitats (Batra , 1966). Seven managed *Apis mellifera* hives were present approximately 250 m from the study area. Observations were conducted at two sub-sites (hereafter referred to as "sites", approximately 5–6 m^2), selected based on high nesting density and clear visibility of nest entrances, which may facilitate nest location via social or environmental cueing (Michener , 2007; Potts and Willmer, 1997). Multi-tunnel nesting aggregations of *H. scabiosae* and *L. malachurum* co-occurred at both sites, as noted in previous studies (Antoine and Forrest , 2021; González et al. , 2018; Polidori et al., 2010). The sites were surrounded by an unmanaged wildflower meadow characterised by *Ranunculus sardous*, *Raphanus raphanistrum*, *Trifolium resupinatum*, *Lysimachia arvensis*, and *Matricaria spp.*, with additional taxa including *Crepis spp.*, *Hordeum spp.*, *Rumex spp.*, and *Taraxacum spp.*. Nesting aggregations were most frequently associated with bare, sandy patches dominated by sparse *Matricaria spp.*, *Trifolium resupinatum*, and *Lysimachia arvensis* vegetation. For the chill-coma recovery assay, samples were collected from sites surrounding the behavioural observation area that were spatially distinct from the observation sites.

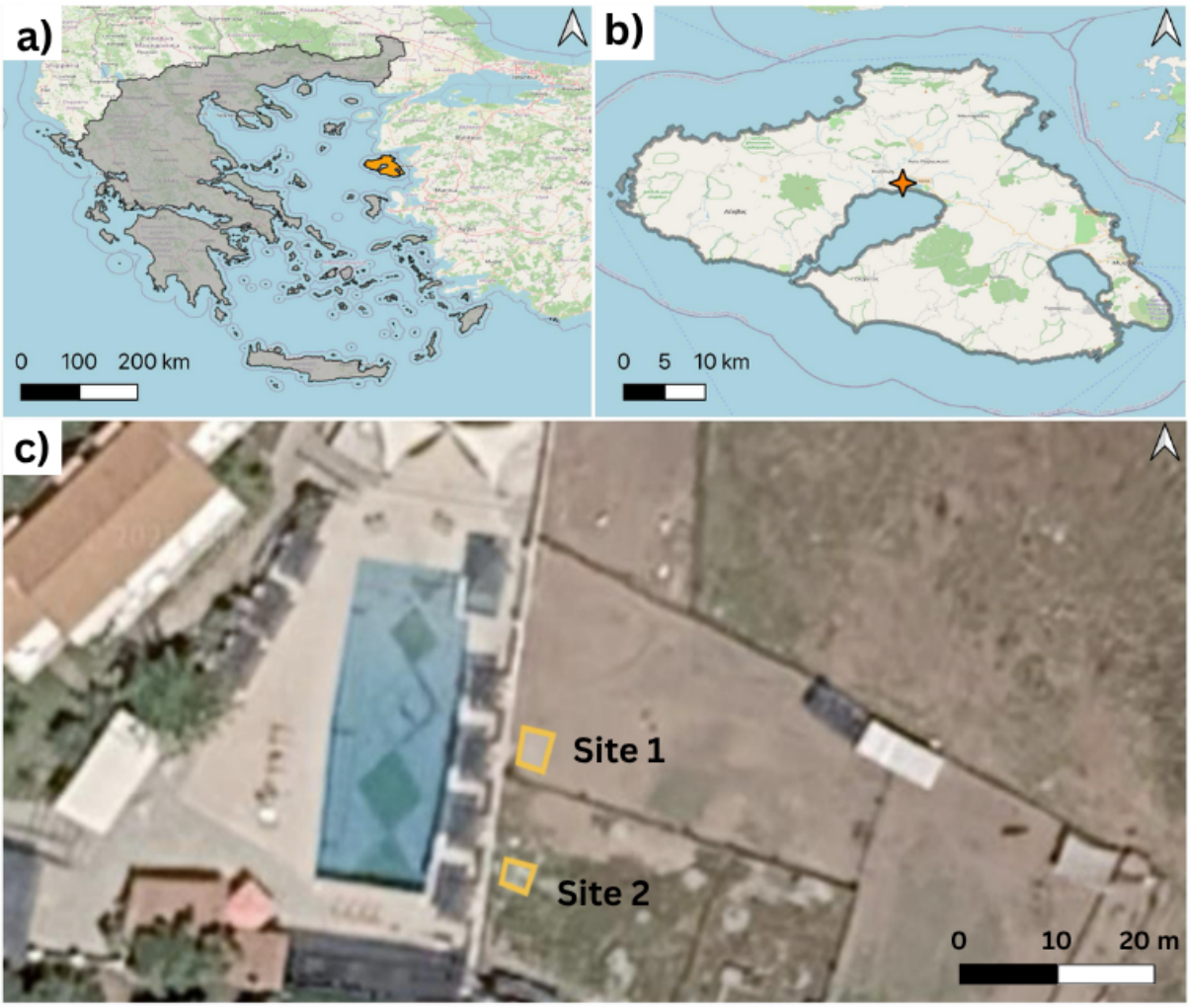


Figure 1: Study location on Lesvos, Greece. (a) Lesvos (orange) within Greece (grey). (b) Skala Kallonis on Lesvos (orange star). Panels (a) and (b) were prepared in QGIS (version 3.40.0, "Bratislava"). (c) Study area showing the approximate locations of Sites 1 and 2 (yellow outlines) in Google Earth satellite imagery dated 24 May 2024, viewed from 107 m above ground level. The midpoint between the sites is at 39°12'32.98"N, 26°12'13.74"E, approximately 1 m above sea level.

2.3 Chill coma recovery assays

To quantify functional thermal thresholds under field-relevant conditions, we measured the temperature and time required for individuals to recover function following cold immobilisation, defined as (i) righting (regaining an upright posture) and (ii) take-off (initiation of flight). Individuals were collected from sites separate from those used for behavioural recordings and placed individually in ventilated plastic vials sealed with cotton (Figure S1). Individuals were not marked for these assays, as the high abundance of bees at collection sites, combined with sampling across multiple sites, reduced the likelihood of repeatedly capturing the same individuals and minimised potential pseudoreplication. Chill coma was induced by cooling individuals in either a freezer or a portable cooling box filled with ice and was confirmed by complete immobilisation and a lack of response to mechanical stimulation by gently flicking the tube (Anderson et al., 2005; David et al., 1998; Oyen et al., 2021). Following immobilisation, individuals were transferred to an experimental arena under ambient conditions and allowed to recover passively. Ambient temperature was continuously recorded using a thermocouple thermometer (Omega HH23, bare type K thermocouple), while bee surface temperature was simultaneously monitored using a thermal imaging camera (FLIR C5) positioned above the experimental arena. Time to right and take-off were recorded using a stopwatch. The experimental arena consisted of a polystyrene enclosure (26 × 17 cm) that helped buffer the bees from external temperature fluctuations and ensure consistent imaging conditions. Body mass was measured prior to trials using a precision balance (Acculab PP2060D; readability 0.001 g), and individuals were handled using fine forceps (Rubis Tech, Switzerland). Body size was estimated using intertegular distance (ITD) as a proxy (Cane , 1987). ITD, defined as the linear distance between the tegulae at the base of the forewings on the dorsal thorax, was measured from dorsal images using ImageJ (Schneider et al., 2012) (Figure S1, Figure S2). Each individual was measured twice from independent images, and the mean value was used for analyses. Measurement error was small relative to the difference between species (mean absolute difference between repeats = 0.019 cm versus a species difference of 0.134 cm; mean coefficient of variation = 4.8%). During recovery, individuals were positioned ventral-side up, and we recorded the elapsed time until each bee regained a normal upright posture (time to right) and its body temperature at that point (righting temperature). We also recorded body temperature at flight initiation (take-off temperature) and the elapsed time until flight initiation (time to take-off).

2.4 Behavioural monitoring and microclimate measurements

Six cameras (a heterogeneous system comprising two action cameras, a mobile phone, an iPad, a camcorder, and a digital single-lens reflex (DSLR) camera) were deployed to record nest-level activity and surrounding arthropod interactions. Two cameras were positioned at Site 1, approximately 1.5 m apart, to capture complementary perspectives: a wide-angle view of the general study area and a zoomed-in view centred on the nest entrances (hereafter referred to as the "focal" view). Three cameras were positioned at Site 2, approximately 1–1.5 m apart, with two capturing focal-views and one providing a wide-angle view. An additional camera was positioned between the two sites to opportunistically capture behavioural observations rather than for systematic sampling and was therefore excluded from the final analysis. One focal-

view camera at Site 2 was also excluded from the final analysis because its recordings were redundant with the other focal-view recordings. Recordings were 10 min long and started at the beginning of each hour. Individual bees were not marked, owing to time constraints and to minimise disturbance during recording. We analysed 166 wide-angle recordings (83 per site, one wide-angle camera per site; one scheduled recording per site was unavailable) and 163 focal-view observations. Focal-view recordings were scheduled hourly (start times 10:00–17:00 on days 1–6 and 09:00–17:00 on days 7–10; sampling was extended to 09:00 from day 7 onwards to capture earlier onset of daily activity), giving 168 scheduled recordings (84 per site). Five were unavailable (days 1, 2 and 6). A separate early-activity check (08:00 on day 7 at Site 1) was not included in the 163 focal-view observations because logger data were unavailable.

Behavioural observations quantified nest-level activity and behavioural repertoires of *H. scabiosae* and *L. malachurum*. Recorded variables included (i) nest entry and exit frequency, and (ii) occurrence and time spent guarding at nest entrances. Nest guarding was quantified only in *H. scabiosae*, as it could not be reliably assessed in *L. malachurum* due to small nesting entrances and restricted camera angles that limited consistent detection of nest-associated behaviour. Consequently, analyses for nest-guarding were restricted to *H. scabiosae* (Figure 6, Supplementary Table S4). Bee abundance was quantified as the number of bees visible outside nests at approximately 2, 4, 6, 8 and 10 min after recording onset (±10 s) and averaged per recording (hence values are multiples of 0.2). In focal-view recordings, counts were made separately for *H. scabiosae* and *L. malachurum*; in wide-angle recordings, the mean number of bees visible outside nests was used as a relative index of activity at the nesting aggregations. Nest entries and exits were counted throughout each 10-min focal-view recording.

Microclimatic conditions were monitored using iButton Thermochron (1-Wire Thermochron; DS1922L; Maxim Integrated, USA) and Hygrochron (1-Wire Hygrochron; DS1923-F5) data loggers deployed across both sites. iButton thermochrons recorded air temperature at 2 m height (within a well-ventilated PVC housing extending beyond a stone wall) and soil temperature at multiple depths (surface, 2–3 cm, 5 cm, and 10 cm). An iButton hygrochron was installed at 10 cm depth near Site 1 to record subsurface humidity and temperature conditions. Nests of *L. malachurum* may reach depths of 10–15 cm during the nest-founding stage (Knerer , 1992; Legewie , 1925; Noll , 1931). Deeper soil measurements (>10 cm) were not conducted in order to minimise repeated disturbance of soil structure, as loggers required periodic retrieval for data verification and maintenance due to battery age and variable reliability. Restricting sampling depth despite nests extending to 20–30 cm depth (Lienhard et al., 2010) reduced disruption to nesting microhabitats and ensured minimal impact on the study system. Light intensity was recorded with a HOBO light logger from 19 April, covering 95 of the 163 focal-view observations (part of day 5 and days 6–10), whereas thermochrons recorded throughout the study. Relative humidity was excluded from analyses due to ceiling effects (predominantly 100% saturation, consistent with previous findings (Potts and Willmer, 1997)), which limited variability and interpretability. Surface temperature records from surface-level loggers at Site 1 and Site 2 were missing for 17–18 April 2026 due to a brief technical error. Air temperature was taken from the 2 m iButton, which recorded throughout the study; a HOBO logger (from 19 April) read on average 4.6 °C lower ($n = 94$; $r = 0.58$), so air temperatures are best interpreted as relative

values.

2.5 Statistical analyses

ITD was compared between species with a linear model. Righting temperature, take-off temperature, time to right and time to take-off were analysed with linear mixed-effects models (LMMs) including ITD, ambient temperature and species in different combinations (and, in sensitivity models for the two temperatures, log-transformed time to right or to take-off), with sampling date as a random intercept. ITD was additionally examined within each species, for each of the four traits, in exploratory linear models that also included ambient temperature (Figure 4); Holm-adjusted p-values were calculated across these eight tests. Wide-angle activity (mean number of bees) was related to air temperature and soil temperature (2–3, 5 and 10 cm) with generalised additive models (GAMs) fitted separately for each layer, with site as a fixed effect and sampling date as a random intercept. Site differences in these relationships were tested by likelihood-ratio tests of shared versus site-specific smooths (sampling date as a random intercept; Holm-adjusted p-values). As robustness checks, models were also fitted without the date term and compared with the date-adjusted models by AIC (maximum likelihood fits), with each temperature centred within sampling date, and with a larger basis dimension ($k = 8$ instead of 5); the share of variance in each temperature layer lying between sampling dates was estimated with a random-intercept linear mixed model (Supplementary Table S5).

Focal-view observations were analysed as follows. Nest-guarding occurrence in *H. scabiosae* was modelled with a binomial generalised linear mixed model (GLMM) and log-transformed guarding duration with an LMM, both with site and day as random intercepts. Because temperature layers were strongly correlated (up to $r \approx 0.9$), each layer was tested in a separate duration model, with Holm-adjusted p-values across the five layers. Surface temperature records were missing for 17–18 April, so the surface-layer duration model used 32 of the 40 recordings with guarding. Abundance was analysed in two ways. First, a linear model of log(abundance + 1) included species, standardised environmental variables (ambient air, soil temperature at three depths, light intensity) and their interactions with species, for the 95 observations with light-logger data; interactions were tested with marginal F-tests and species-specific slopes estimated with `emmeans`. As a sensitivity analysis, the model was refitted with random intercepts for day and observation. Second, to test whether species differed in timing beyond temperature, LMMs of log-transformed abundance included species, hour of day, air temperature, shallow soil temperature, site and species interactions, with day and observation as random intercepts (the latter accounts for both species being recorded in each observation). Nest entries and exits were analysed as counts with negative binomial GLMMs (`glmmTMB`) using the same fixed effects and random intercepts. These models used all 163 observations and were repeated with light added for the 95 observations with light data (for nest exits in this subset, the null model failed to converge and was excluded from the comparison). Models were fitted by maximum likelihood and ranked by AICc, and collinearity was checked with variance inflation factors. Daily activity curves were estimated with species-specific GAM smooths of hour, compared with a shared smooth by likelihood-ratio test, and centre of activity (abundance-weighted mean hour) and first active hour were compared between species with LMMs. Nest entry and exit frequencies were

also modelled, as an exploratory analysis, with Gaussian GAMs of log(count + 1) including smooths of each thermal layer and light (95 observations).

All analyses were performed in R (version 4.4.2; R Core Team, 2024) using RStudio (version 2024.9.1+394; Posit Team, 2024; "Cranberry Hibiscus" release). Mixed models were fitted with `lme4` and `lmerTest`, count models with `glmmTMB`, GAMs with `mgcv`, slopes with `emmeans`, model comparison with `MuMIn`, and collinearity with `car` and `performance`. Residuals of the wide-angle and focal-view GAMs were checked with `DHARMa` (Supplementary Table S6). Figures were produced with `ggplot2` and `patchwork`.

3 Results

3.1 Interspecific differences in body size

Body size differed between species, with *H. scabiosae* exhibiting a larger ITD than *L. malachurum* (Figure 2, Supplementary Table S7). Mean ITD was 0.340 ± 0.0329 cm in *H. scabiosae* and 0.206 ± 0.0220 cm in *L. malachurum*, corresponding to a 1.65-fold difference in body size between species. A linear model confirmed a significant effect of species identity on ITD, with *L. malachurum* exhibiting a smaller ITD than *H. scabiosae* ($\beta = -0.134$ cm, SE = 0.0095, $t = -14.08$, $p < 0.001$). Body mass and ITD were strongly correlated ($r = 0.75$), indicating substantial collinearity between body size metrics. Body mass was therefore excluded from subsequent analyses, and ITD retained as the primary proxy for body size because it is less affected by feeding state.

3.2 Recovery temperature

3.2.1 Righting temperature

A linear mixed-effects model indicated a significant positive effect of ambient temperature on righting temperature ($\beta = 0.98$, SE = 0.15, $t = 6.55$, $p < 0.001$), indicating that bees righted at higher body temperatures under warmer environmental conditions. Species identity also had a significant effect, with *L. malachurum* exhibiting lower righting temperatures than *H. scabiosae* ($\beta = -2.49$, SE = 0.88, $t = -2.84$, $p = 0.0078$), suggesting species-specific differences in acute recovery temperature thresholds (Figure 3). ITD was not included in this final model. The association with ambient temperature remained after accounting for time to right ($\beta = 0.98$, SE = 0.15, $t = 6.49$, $p < 0.001$), but the species difference was weaker in this model ($\beta = -1.85$, SE = 0.93, $t = -1.99$, $p = 0.056$); time to right was not associated with righting temperature ($p = 0.27$). Variation among sampling dates remained (variance = 1.148), indicating between-day heterogeneity in righting temperature.

3.2.2 Take-off temperature

A linear mixed-effects model indicated a strong positive effect of ambient temperature on take-off temperature ($\beta = 1.41$, SE = 0.23, $t = 6.02$, $p < 0.001$), indicating that bees initiated flight at higher body temperatures under warmer environmental conditions. Species identity showed a negative marginal effect, with *L. malachurum* tending to initiate flight at slightly

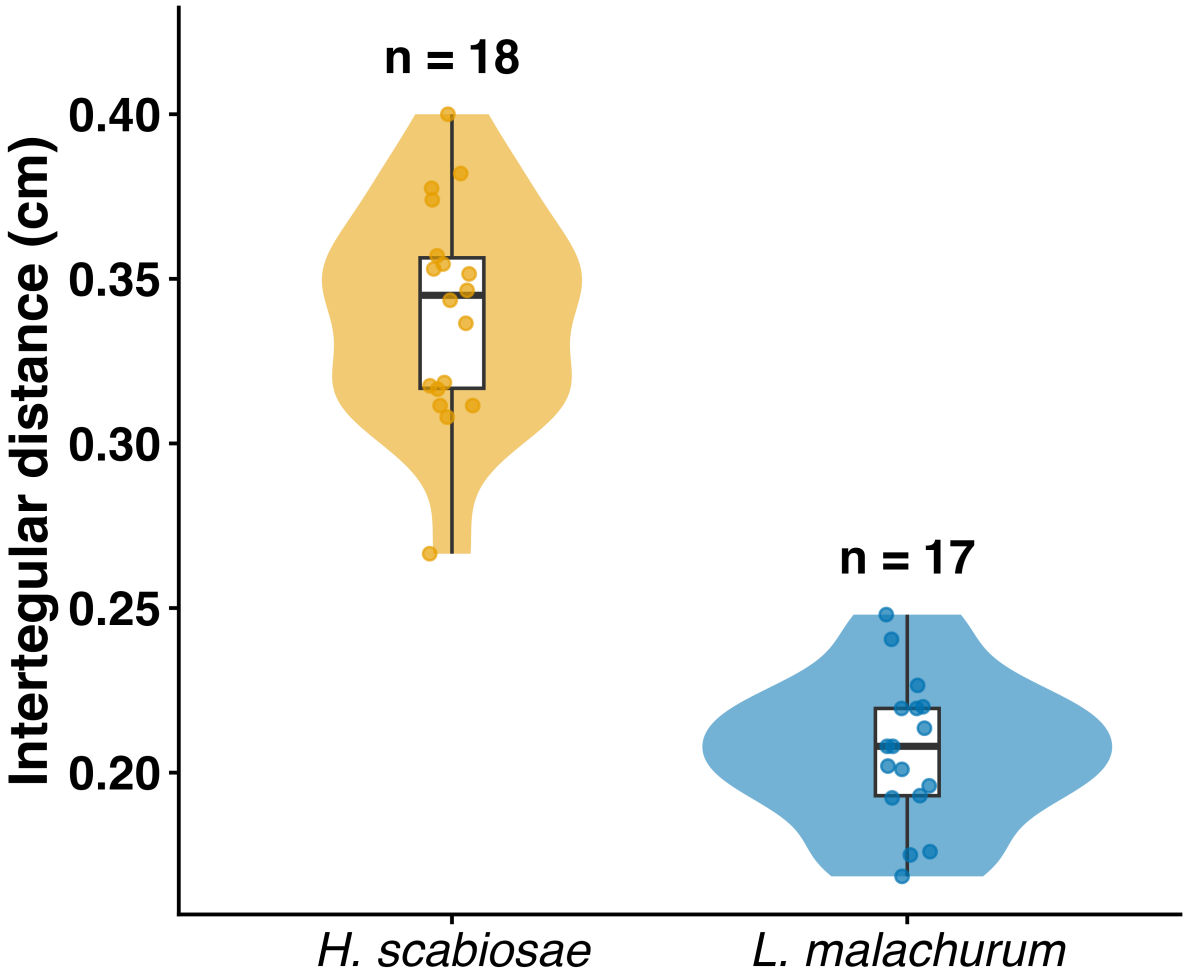


Figure 2: Intertegular distance (ITD) in *Halictus scabiosae* and *Lasioglossum malachurum*. Violin plots show the distributions for each species. Boxplots indicate the median and interquartile range, with whiskers extending to the most extreme observations within 1.5× the interquartile range. Points represent individual bees, and sample sizes (n) are shown above each group.

lower temperatures than *H. scabiosae*, although this difference was not statistically supported ($\beta = -2.10$, SE = 1.12, $t = -1.88$, $p = 0.069$). The association with ambient temperature remained after accounting for time to take-off ($\beta = 1.29$, SE = 0.23, $t = 5.51$, $p < 0.001$), and take-off temperature also increased with time to take-off ($\beta = 2.21$, SE = 0.70, $t = 3.16$, $p = 0.004$); the species difference remained unsupported ($p = 0.088$).

3.3 Recovery time

3.3.1 Time to right

Because ITD was collinear with species identity, the duration models testing species differences included species identity and ambient temperature but not ITD. Time to right (log-transformed) tended to be shorter in *L. malachurum*, but the species difference was not statistically supported after accounting for ambient temperature ($\beta = -0.42$, SE = 0.21, $t = -1.98$, $p = 0.057$). Ambient temperature was not associated with time to right ($\beta = 0.033$, SE = 0.036, $t = 0.90$, $p = 0.377$). Variation among sampling dates was low (variance = 0.066).

3.3.2 Time to take-off

Neither species identity nor ambient temperature was associated with time to take-off (log-transformed). The species effect was not statistically supported ($\beta = 0.003$, SE = 0.251, $t = 0.01$, $p = 0.989$), and ambient temperature had no detectable effect ($\beta = 0.058$, SE = 0.053, $t = 1.09$, $p = 0.284$). The estimated variance among sampling dates was zero, and the model had a singular fit. Refer to Supplementary Table S8 for model statistics.

3.4 Body size and recovery within species

In exploratory species-specific linear models that also included ambient temperature, ITD was negatively associated with righting temperature ($\beta = -93.07$ per cm, SE = 36.45, $t = -2.55$, $p = 0.023$) and take-off temperature ($\beta = -91.60$ per cm, SE = 41.59, $t = -2.20$, $p = 0.045$) in *L. malachurum* ($n = 17$), but not in *H. scabiosae* (righting temperature: $p = 0.26$; take-off temperature: $p = 0.67$; $n = 18$). ITD was not associated with time to right or time to take-off in either species (all $p \geq 0.24$; Figure 4). Neither temperature association remained significant after Holm correction across the eight tests (smallest adjusted $p = 0.18$), so we treat them as exploratory (Supplementary Table S8).

3.5 Thermal correlates of bee activity

Wide-angle activity was analysed for 166 recordings on 10 days (15–24 April; Supplementary Table S9). In generalised additive models including site and a random intercept for sampling date, activity varied with ambient air temperature (edf = 2.95, $F = 22.63$, $p < 0.001$; adjusted $R^2 = 0.43$), shallow soil temperature at 2–3 cm (edf = 3.48, $F = 11.72$, $p < 0.001$; adjusted $R^2 = 0.34$), middle soil temperature at 5 cm (edf = 3.64, $F = 13.26$, $p < 0.001$; adjusted $R^2 = 0.36$), and deep soil temperature at 10 cm (edf = 3.83, $F = 38.11$, $p < 0.001$; adjusted $R^2 = 0.57$; Figure 5, Supplementary Table S10). Adjusted R^2 values include the date effect. DHARMa residual checks showed no evidence of departures from uniformity, dispersion, or outliers (Supplementary

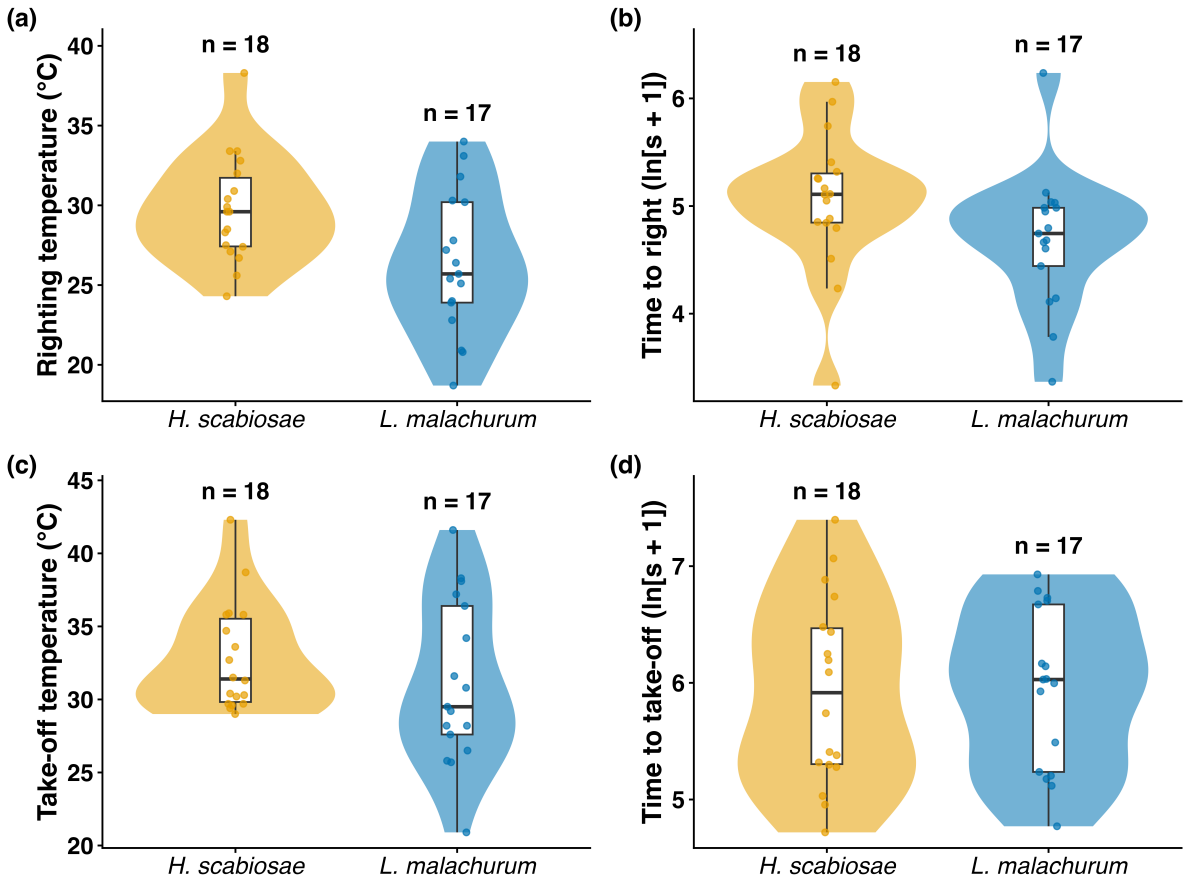


Figure 3: Thermal performance traits in *H. scabiosae* and *L. malachurum*: (a) righting temperature, (b) time to right, (c) take-off temperature, and (d) time to take-off. Violin plots show the distributions for each species. Boxplots indicate the median and interquartile range, with whiskers extending to the most extreme observations within 1.5× the interquartile range. Points represent individual bees, and sample sizes (n) are shown above each group. Time measures are plotted as $\ln(\text{seconds} + 1)$.

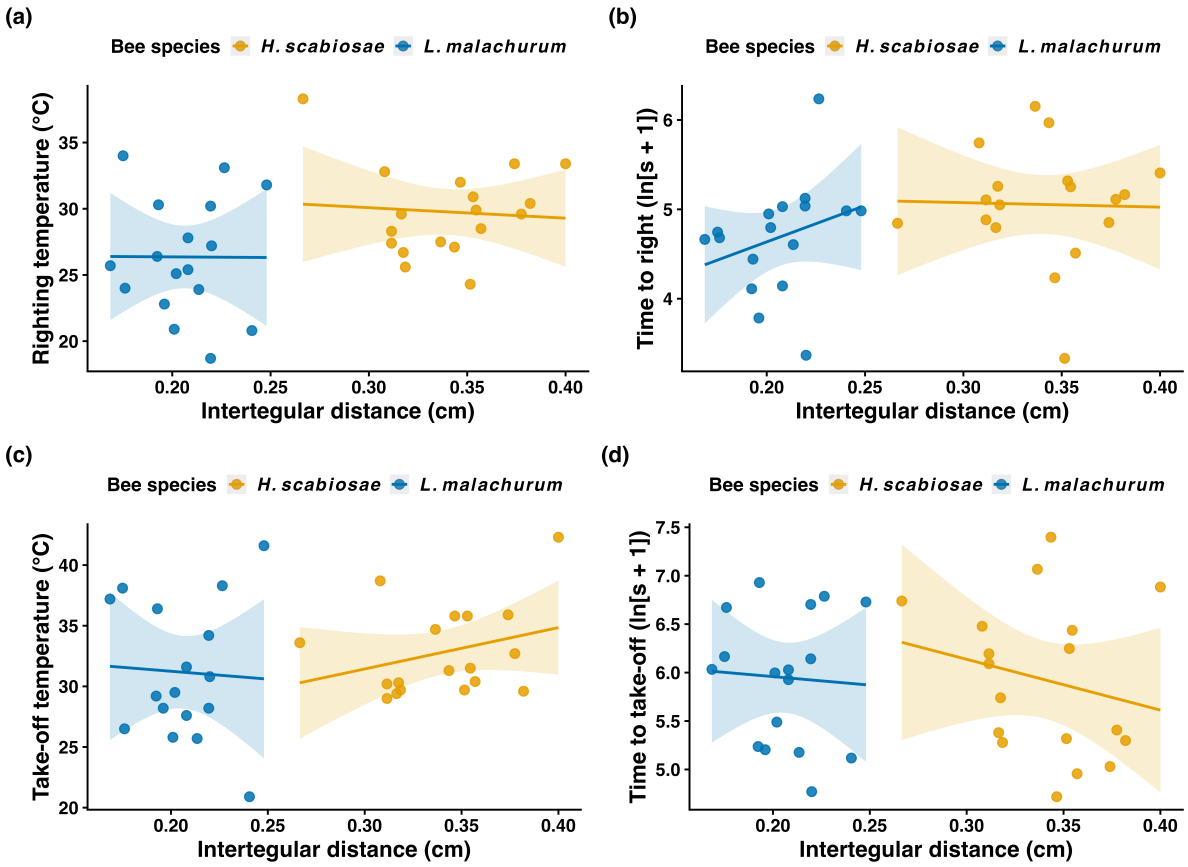


Figure 4: Relationships between intertegular distance (ITD) and (a) righting temperature, (b) time to right, (c) take-off temperature, and (d) time to take-off in *Halictus scabiosae* and *Lasioglossum malachurum*. Points show individual bees; lines show species-specific linear fits, with shaded bands indicating 95% confidence intervals. Time measures are plotted as ln(seconds + 1).

Table S6), but the low k-index values (0.69–0.73) indicated residual structure; pooled curves are therefore descriptive. Site-specific models showed that the relationships with shallow and middle soil temperature differed between sites (shallow: $\chi^2 = 15.19$, df = 1.03, Holm-adjusted $p = 0.0002$; middle: $\chi^2 = 32.68$, df = 4.35, Holm-adjusted $p < 0.001$; Holm adjustment across the three valid comparisons; Supplementary Figure S3), whereas the relationship with deep soil temperature did not ($\chi^2 = 1.85$, df = 4.22, Holm-adjusted $p = 0.79$). For shallow soil, activity was associated with temperature at Site 2 ($p < 0.001$), but not at Site 1 ($p = 0.857$). Within sites, air-temperature smooths were linear (edf = 1.0), and a formal site comparison was not valid because the site-specific smooths used fewer effective degrees of freedom than the shared smooth.

These results were robust to how sampling date and basis dimension were handled (Supplementary Table S5). Including sampling date improved model fit for every layer (ΔAIC = 37–93), and most of the variation in shallow (72%) and middle (75%) soil temperature occurred within days, compared with 59% for air temperature and 96% for deep soil. Without the date term, shallow soil temperature was only weakly related to activity ($p = 0.056$, with site as a fixed effect), whereas all four layers were clearly related to activity when temperature was centred within sampling date ($p < 0.001$ in each case). All four relationships remained significant with a larger basis dimension ($k = 8$).

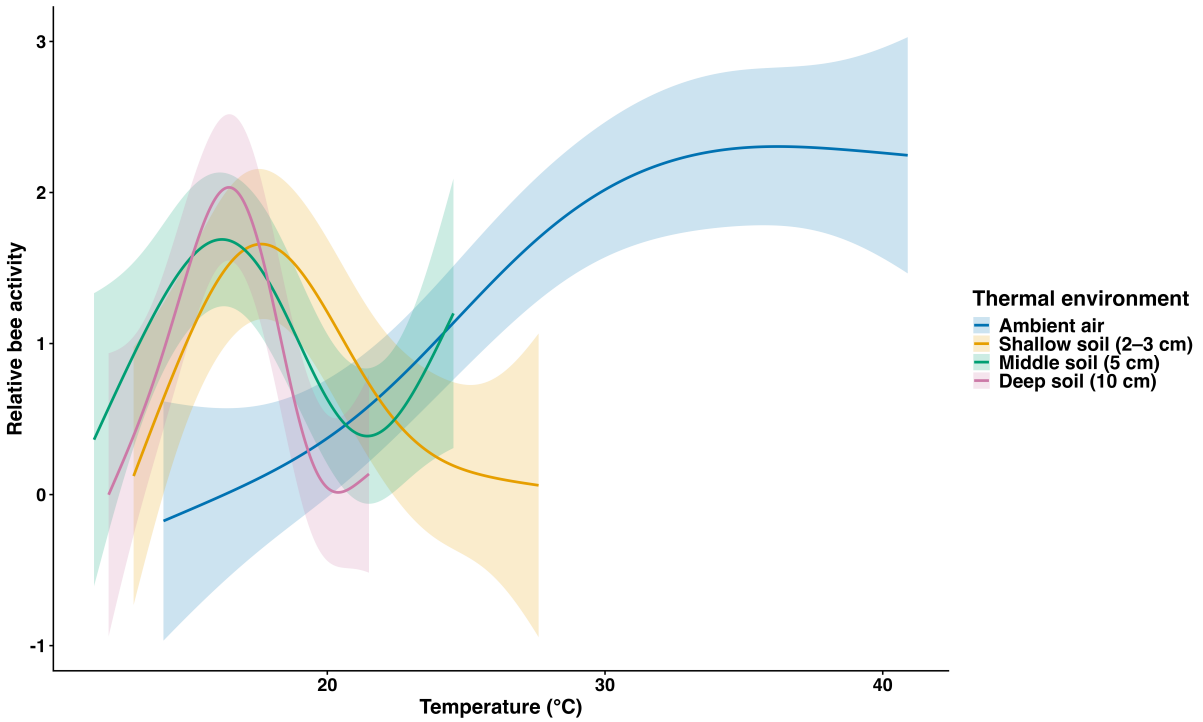


Figure 5: Generalised additive model (GAM) fits showing wide-angle bee activity (mean number of bees observed) in relation to (a) ambient air temperature, (b) shallow soil temperature (2–3 cm), (c) middle soil temperature (5 cm), and (d) deep soil temperature (10 cm). Soil temperatures were averaged across sensors at the two sites. Each thermal variable was fitted in a separate model with site as a fixed effect and sampling date as a random intercept. Curves show predictions for Site 1 with the date effect excluded. Models used 166 recordings collected from 15–24 April.

3.6 Nest guarding duration and probability of nest guarding in *H. scabiosae*

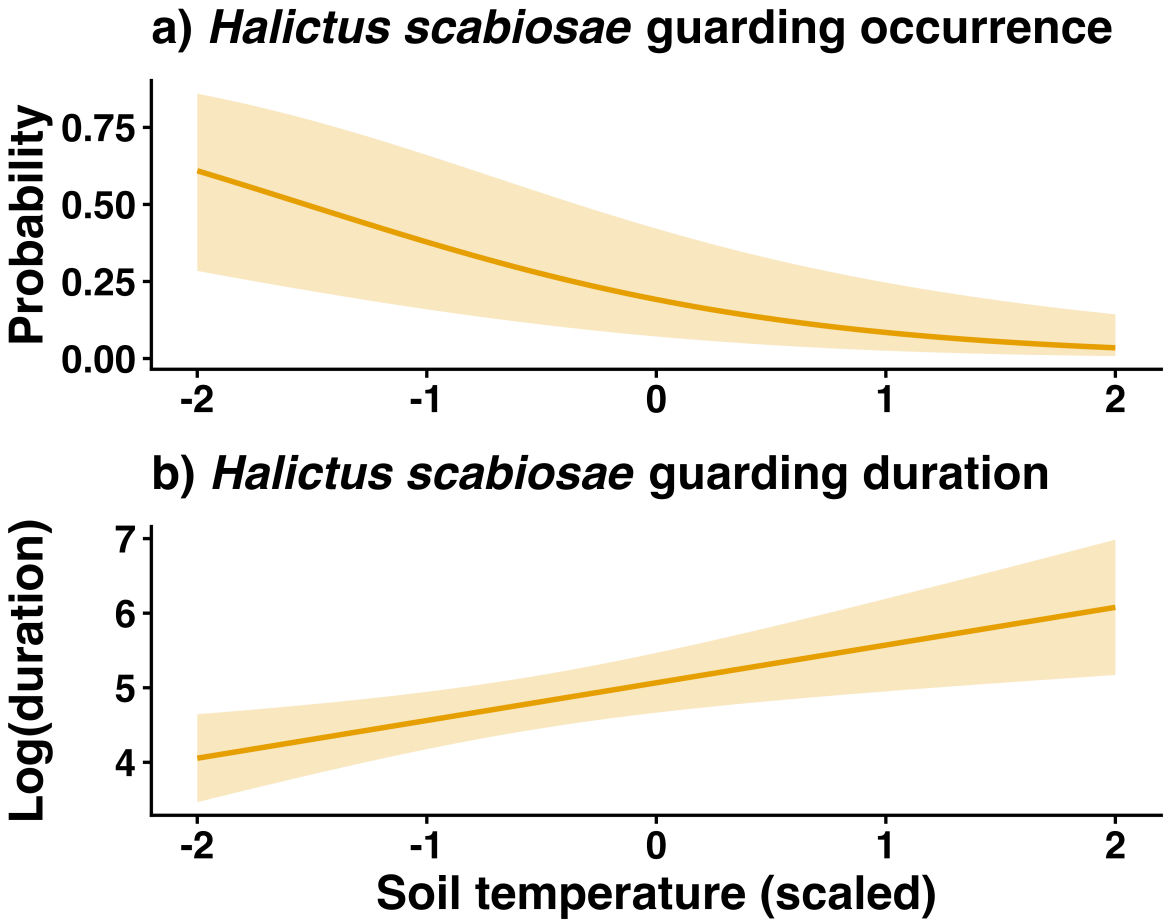


Figure 6: Effects of subsurface soil temperature on nest guarding in *H. scabiosae*. (a) Predicted probability of guarding in relation to shallow soil temperature (2–3 cm), from a binomial generalised linear mixed-effects model. (b) Predicted log-transformed guarding duration in relation to middle soil temperature (5 cm), from a linear mixed-effects model. Temperatures were standardised using the mean and standard deviation (SD) across 163 focal-view observations (shallow: 19.65 ± 3.20 °C; middle: 18.51 ± 2.83 °C). Values of -2 and $+2$ correspond to mean ± 2 SD (shallow: 13.26–26.05 °C; middle: 12.84–24.18 °C). Observed ranges were 13.0–27.6 °C for shallow soil and 11.6–24.5 °C for middle soil. Lines show model predictions; shaded ribbons indicate 95% confidence intervals.

Nest guarding duration increased with soil temperature at 5 cm depth ($\beta = 0.506 \pm 0.166$, $t = 3.04$, $p = 0.004$; $n = 40$ recordings with guarding; Supplementary Table S11). Because the temperature layers were strongly correlated, each was tested in a separate model; after Holm correction, only soil temperature at 5 cm depth remained significant (adjusted $p = 0.022$), although deep soil (10 cm) showed a similar positive association (adjusted $p = 0.074$). Ambient air, surface and shallow soil temperatures were not significant predictors. Random effects indicated moderate variation among days and low variation among sites.

In contrast, the probability of nest guarding decreased significantly with increasing shallow soil temperature (2–3 cm depth; $\beta = -0.94 \pm 0.27$, $z = -3.50$, $p < 0.001$; $n = 163$ observations, of which 40 included guarding), indicating reduced guarding activity under warmer subsurface

conditions (Supplementary Table S11). Variation among days and sites was moderate.

3.7 Species differences in abundance, timing and thermal dependence of activity

Focal-view observations comprised 163 observations over 10 days (hourly, 09:00–17:00; 09:00 was sampled on four days only). Temperature was recorded for all observations, whereas the light logger recorded for 95 observations (part of day 5 and days 6–10). Analyses without light therefore used all 163 observations, and analyses including light used the 95 observations with light data.

3.7.1 Time of day and temperature

Abundance was substantially better supported by a model including hour, ambient air temperature and shallow soil temperature than by models including only time of day or only temperature (ΔAICc = 38.5 and 31.5, respectively). *L. malachurum* was less abundant than *H. scabiosae* ($\beta = -0.34 \pm 0.03$, $p < 0.001$). In *H. scabiosae*, abundance increased with air temperature (0.16 ± 0.03 per SD, $p < 0.001$) and shallow soil temperature (0.19 ± 0.04, $p < 0.001$) and decreased with hour of day after accounting for temperature (-0.10 ± 0.02 per hour, $p < 0.001$). In *L. malachurum*, none of these relationships differed clearly from zero (all $p \geq 0.06$). The species differed significantly in all three relationships (species × hour, $p < 0.001$; species × air temperature, $p = 0.012$; species × shallow soil temperature, $p < 0.001$).

Nest entries were best supported by the model including hour and temperature (Akaike weight 0.57), although the time-only model was nearly equivalent (ΔAICc = 0.65). Entries of *H. scabiosae* decreased with hour ($\beta = -0.39 \pm 0.15$, $p = 0.008$) and increased with air temperature (0.59 ± 0.24, $p = 0.015$). Entries of *L. malachurum* also decreased with hour (-0.43 ± 0.18, $p = 0.020$) but did not vary with air temperature (0.22 ± 0.26, $p = 0.40$); the species did not differ significantly in their air-temperature response (species × air temperature, $p = 0.13$). Nest exits were best supported by the time-only model (Akaike weight 0.91). Exits decreased with hour in both species (*H. scabiosae*: -0.38 ± 0.15, $p = 0.012$; *L. malachurum*: -0.52 ± 0.20, $p = 0.011$), with no effect of air or shallow soil temperature in either species (all $p \geq 0.17$) and no species differences (all species interactions $p \geq 0.14$).

3.7.2 Daily activity curves

Species-specific smooths of hour fitted better than a shared smooth ($p = 0.016$), but both species peaked around midday (predicted peak at 11.6 h in *H. scabiosae* and 11.8 h in *L. malachurum*; Figure 7). The centre of activity did not differ between species (0.01 ± 0.25 h, $p = 0.96$). *L. malachurum* was first detected 0.60 ± 0.28 h later than *H. scabiosae* ($p = 0.041$), a difference confounded with its lower abundance and with the 10:00 start of observations on days 1–6.

3.7.3 Effect of light intensity

In the 95 observations with light data, the light-only model was best supported for abundance (Akaike weight 0.49), nest entries (0.86) and, for nest exits, the light-only model was best sup-

ported (next-best model ΔAICc = 1.85; the null model failed to converge and was excluded). Models without light were less supported for abundance (ΔAICc $\geq$ 4.5). Light intensity was correlated with ambient air temperature ($r = 0.77$) and hour ($r = -0.58$), so their effects cannot be fully separated. When all terms were included, effects of hour were not significant in either species for entries or exits (all $p \geq 0.058$) and no species interaction was significant for nest entries or exits (all $p \geq 0.07$). For nest entries, the light slope was clearer in *L. malachurum* (1.82 ± 0.68, $p = 0.008$) than in *H. scabiosae* (0.64 ± 0.37, $p = 0.088$), although the species difference was not supported ($p = 0.075$).

3.7.4 Abundance across thermal layers

A linear model including all thermal layers and light (95 observations, each contributing one row per species; $R^2 = 0.556$; $F_{11,178} = 20.3$, $p < 0.001$) indicated that abundance of *H. scabiosae* increased with shallow soil temperature ($\beta = 0.32 \pm 0.08$, $p < 0.001$) and light intensity (0.24 ± 0.05, $p < 0.001$) and decreased with soil temperature at 5 cm (-0.29 ± 0.10, $p = 0.006$), whereas ambient air and deep soil temperature had no detectable effect ($p = 0.61$ and 0.56). In *L. malachurum*, no slope differed from zero (all $p \geq 0.24$). Only the species $\times$ light interaction was statistically supported ($F_{1,178} = 8.24$, $p = 0.005$); the interaction with shallow soil temperature was marginal ($p = 0.058$), and those with air, 5 cm and 10 cm soil temperature were not ($p = 0.27$, 0.25 and 0.42; Supplementary Table S11). The opposing signs of the shallow and 5 cm soil terms reflect collinearity among soil layers and indicate relative rather than independent associations (Figure 8, Supplementary Figure S4). A sensitivity analysis with random intercepts for day and observation gave the same species-specific slopes (day variance was estimated at zero, giving a singular fit). The species $\times$ light interaction remained supported ($p = 0.002$), and the species $\times$ shallow soil interaction, marginal in the linear model, was supported in this model ($p = 0.037$). As individual bees could not be distinguished, pseudoreplication cannot be fully excluded.

3.8 Environmental correlates of nest entry and exit frequency

The focal-view Gaussian GAMs of nest entry and exit frequency are exploratory, as residual diagnostics indicated departures from model assumptions, particularly for *L. malachurum* (Supplementary Table S6); associations not also supported by the count models are not interpreted. Nest entries of both species were non-linearly related to surface soil temperature (*H. scabiosae*: edf = 2.13, $F = 3.54$, $p = 0.024$; *L. malachurum*: edf = 2.36, $F = 6.32$, $p < 0.001$), whereas nest exits were associated with surface and shallow soil temperature in *H. scabiosae* and with air temperature in *L. malachurum* (Supplementary Table S12; Supplementary Figures S5). Light and deep soil temperature were not significant in any model, and the models explained 23.0–40.5% of deviance.

4 Discussion

In this study, we investigated species-specific patterns of abundance and behavioural responses to environmental variation in temperature and light intensity, alongside thermal performance measured via chill coma recovery, in two sweat bee species, *H. scabiosae* and *L. malachurum*,

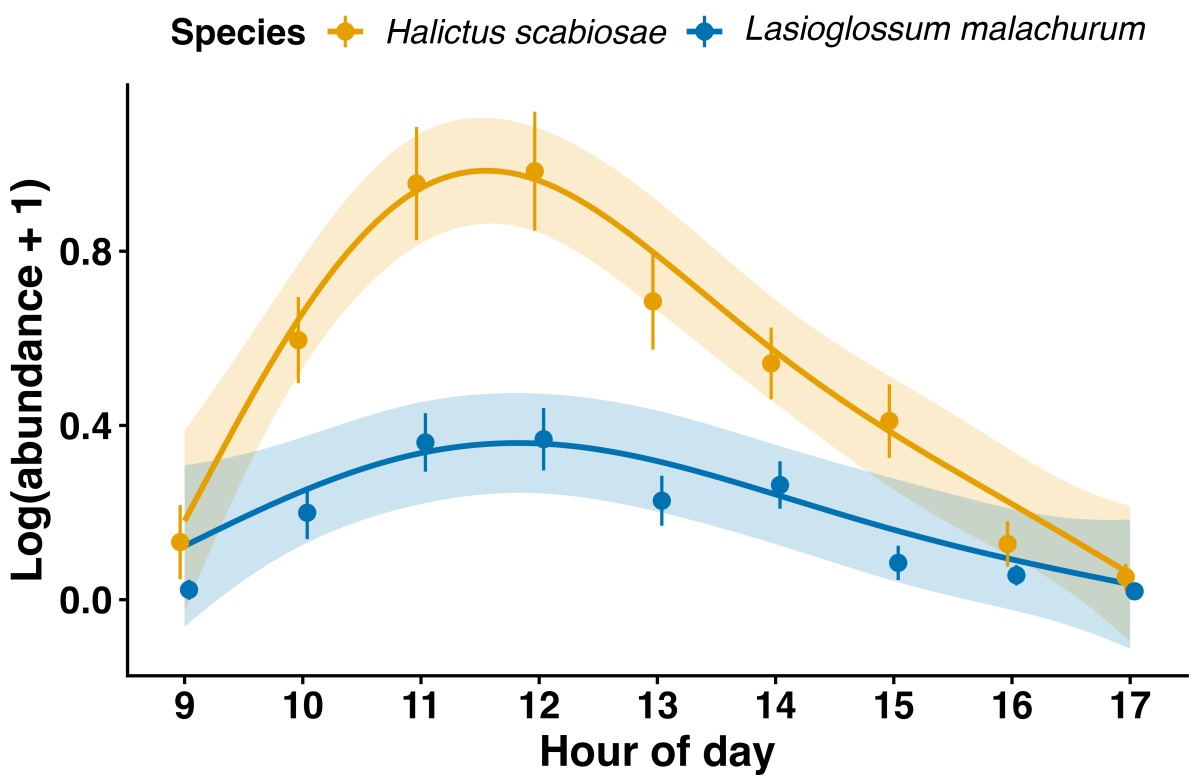


Figure 7: Hourly activity in focal-view observations of *H. scabiosae* and *L. malachurum*. Points show the mean ± standard error of log(abundance + 1) at each hour. Lines show species-specific smooths from generalised additive models (GAMs), with random effects excluded; shaded bands indicate 95% confidence intervals.

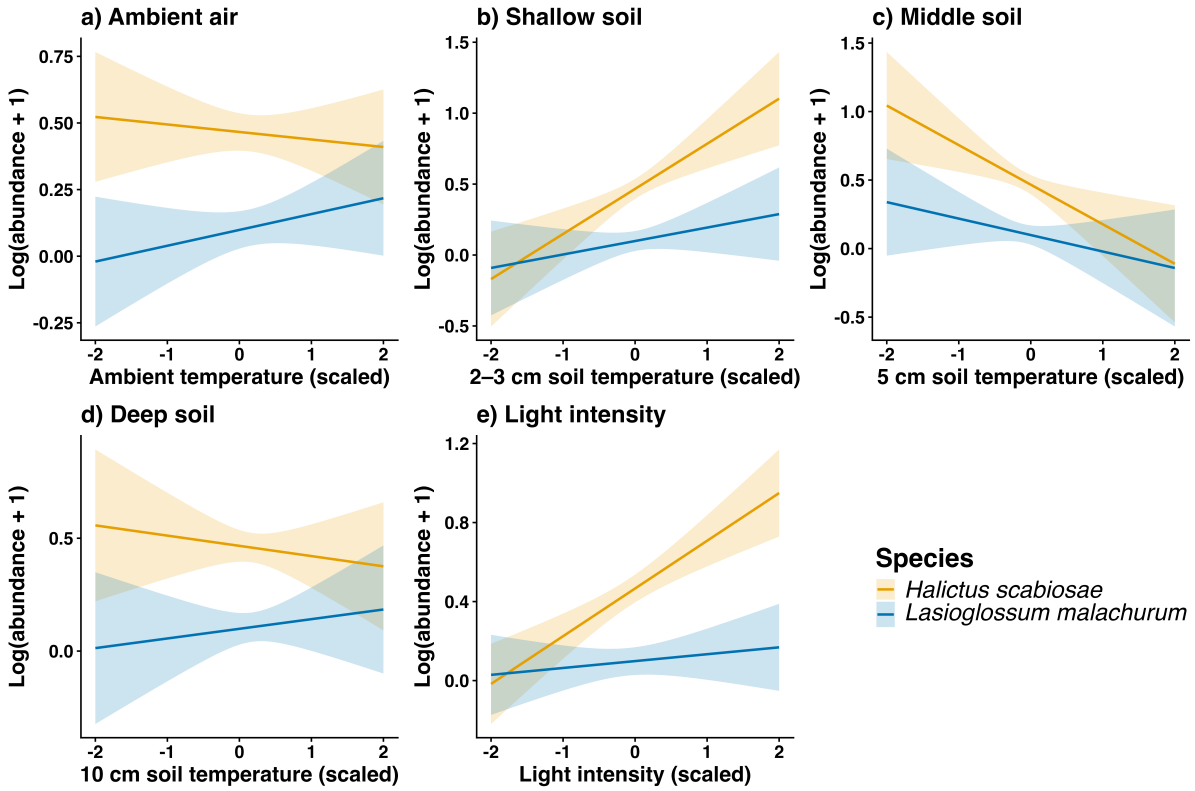


Figure 8: Relationships between environmental variables and bee abundance in *H. scabiosae* and *L. malachurum*, predicted from a linear model fitted to 95 focal-view observations with light data. Panels show predicted abundance across (a) ambient air temperature, (b) shallow soil temperature (2–3 cm), (c) middle soil temperature (5 cm), (d) deep soil temperature (10 cm), and (e) light intensity. Temperature variables were standardised using the mean and standard deviation across all 163 focal-view observations; light intensity was standardised using the 95 observations with light data. Soil temperatures had the following mean ± standard deviation: shallow, 19.65 ± 3.20 °C; middle, 18.51 ± 2.83 °C; deep, 17.62 ± 2.17 °C. Values of −2 and +2 therefore correspond to mean ±2 standard deviations (shallow: 13.26–26.05 °C; middle: 12.84–24.18 °C; deep: 13.27–21.96 °C). Observed ranges were 13.0–27.6 °C for shallow soil, 11.6–24.5 °C for middle soil, and 12.1–21.5 °C for deep soil. Lines show model predictions; shaded ribbons indicate 95% confidence intervals.

on Lesvos island, Greece. Body size differed significantly between species, with *H. scabiosae* exhibiting larger ITD than *L. malachurum*.

4.1 Chill-coma recovery assays

Chill-coma recovery assays revealed a consistent association between ambient temperature and both righting and take-off temperatures in *H. scabiosae* and *L. malachurum*. Bees righted and initiated flight at higher body temperatures under warmer ambient conditions, and these associations remained after accounting for time to right and time to take-off, respectively. Take-off temperature also increased with time to take-off, consistent with bees warming passively until flight became possible. Species identity was associated with righting temperature, with *L. malachurum* righting at lower temperatures than *H. scabiosae*, although this difference was weaker when time to right was included in the model. Because ITD was collinear with species identity, it was excluded from the final righting-temperature model. In exploratory species-specific models that accounted for ambient temperature, ITD was not associated with righting or take-off temperature in *H. scabiosae* but was negatively associated with both in *L. malachurum*, although neither association survived correction for multiple tests. Neither time to right nor time to take-off showed a statistically supported association with species identity or ambient temperature; ITD was not included in these duration models because of its collinearity with species identity. Variation among sampling dates was present for righting temperature but negligible for take-off temperature and both duration measures. Overall, thermal thresholds were associated with ambient temperature and showed some species differentiation, whereas the duration measures showed weaker evidence of species or temperature effects.

4.2 Species-specific behavioural responses to microclimatic variation

Overall activity at the nesting aggregations was associated with air temperature and with soil temperature at all three depths once site and sampling date were accounted for (Figure 5), although the shallow and middle soil relationships differed between sites (Supplementary Figure S3). *H. scabiosae* was substantially more abundant than *L. malachurum*, but this may partly reflect differences in nest detectability and local nest density, as its nests were larger and more conspicuous. Activity of *H. scabiosae* was structured by thermal conditions, light and time of day: abundance increased with warmer air and shallow soil (all 163 observations) and, in the subset with light data, with shallow soil temperature and light intensity, and, once temperature was accounted for, declined through the day. Warmer conditions may facilitate foraging by relaxing thermoregulatory constraints on flight (Lienhard et al., 2010; Willmer and Stone , 2004), whereas the afternoon decline may reflect resource availability or the timing of foraging bouts, which these data cannot distinguish. In contrast, abundance of *L. malachurum* showed weak or no relationships with temperature, light or hour, and differed significantly from *H. scabiosae* in its response to temperature, hour and light. Abundance of *L. malachurum* was therefore less tightly coupled to temperature than that of *H. scabiosae*, although its lower counts may have limited power to detect slopes.

Nest entries and exits (counts of bees entering and leaving nest entrances, used as indices of nest-associated traffic that includes foraging trips but cannot distinguish them from other

movements) did not show a clear species contrast in count models. Entries and exits of both species declined through the day, entries of *H. scabiosae* increased with air temperature while those of *L. malachurum* did not (species difference $p = 0.13$), and exits were explained best by time of day alone. In the subset with light data, the light-only model was best supported for nest entries. The estimated light slope was larger for *L. malachurum* than for *H. scabiosae*, although the species difference was not statistically supported ($p = 0.075$). Light covaries with air temperature and hour, so their effects cannot be fully separated. Associations of nest entries and exits with surface and soil temperature in the generalised additive models were not supported by the count models, so we do not interpret them as evidence of an effect.

Both species peaked around midday and their centres of activity were indistinguishable, although the shape of their daily curves differed ($p = 0.016$), so differences in activity lay in the strength of environmental dependence rather than in the timing of daily activity.

Nest guarding in *H. scabiosae* showed two distinct responses: the probability of guarding decreased with increasing shallow soil temperature, whereas guarding duration increased with soil temperature at 5 cm depth, the layer that remained best supported after correcting for the five layers tested, suggesting that once guarding began, warmer nest-adjacent conditions were associated with longer guarding durations. Because soil temperatures across depths are correlated, coefficients indicate relative associations rather than independent thermal effects.

4.3 Linking thermal performance to activity under changing spring conditions

L. malachurum righted at lower body temperatures than *H. scabiosae*, which could indicate greater cold tolerance. However, righting and take-off temperatures rose with ambient temperature (slopes $\approx$ 1–1.4), indicating that bees warmed passively towards ambient conditions and that these are not fixed thresholds; species and body size are also confounded, and the duration measures did not differ between species. We therefore treat greater cold resilience of *L. malachurum* as a hypothesis. Greater righting-temperature tolerance did not translate into earlier activity: both species peaked around midday, their centres of activity were indistinguishable, and *L. malachurum* was first detected later, a difference confounded with its lower abundance and with the 10:00 start of observations on days 1–6. The species differed instead in how strongly activity tracked environmental conditions. Abundance of *H. scabiosae*, the species with the higher righting temperature, increased with shallow soil temperature and light, whereas that of *L. malachurum* showed weak or no relationships; this species difference was supported for abundance but not for nest entries and exits. This is in the direction predicted by our hypothesis that the less cold-resilient species sustains higher activity at higher temperatures, although behavioural persistence could not be compared because nest guarding was recorded only in *H. scabiosae*. Because *H. scabiosae* was 1.65-fold larger in ITD, body size and species identity are largely confounded, so the species difference in righting temperature cannot be attributed to intrinsic physiology rather than size. Within species, ITD was unrelated to righting and take-off temperature in *H. scabiosae* but negatively associated with both in *L. malachurum*, suggesting, as an exploratory result, that size may contribute. The coupling of *H. scabiosae* to soil temperature, together with site-dependent soil relationships, suggests that ground-level conditions vary

over small spatial scales and may matter at least as much as air temperature for ground-nesting bees. If warmer conditions persist, activity of the two species may respond differently, which future multi-season work could test. These inferences are based on a single season at two sites and should be treated as hypotheses.

4.4 Thermal ecology and nesting behaviour in halictid bees

Halictidae display substantial variation in social organisation within and among species (Hirata et al., 2005; Packer, 1998; Plateaux-Quénu , 1993; Richards et al., 2003; Yanega, 1988), which is often associated with environmental conditions (Eickwort et al., 1996; Miyanaga et al., 1999; Yanega, 1997) and, in some cases, genetic influences on social traits (Danforth et al., 2003; Plateaux-Quénu et al., 2000; Soucy, 2002). Ground-nesting insects are more strongly influenced by soil temperature than air temperature in relation to nest-site selection, daily activity patterns, and energetic allocation to foraging and life-history processes (Cameron et al., 1996; Crist and Williams , 1999; Potts and Willmer, 1997; Strohm and Linsenmair, 1998; Vogt et al., 2003; Weissel et al., 2006). In *L. malachurum*, increasing soil temperatures reduce brood development time and shorten activity pauses between broods, likely due to physiological effects on larval development (Weissel et al., 2006). Higher activity at warmer temperatures reported previously for *L. malachurum* (Weissel et al., 2006) is consistent with our wide-angle results, in which overall activity increased with air and soil temperature, but not with the weak species-level relationships we found for *L. malachurum*, which may reflect its low counts. For *H. scabiosae*, nesting preferences could be examined in a similar framework to *Halictus rubicundus*, which shows higher nesting densities in areas with warmer soil temperatures at 5 cm depth relative to surrounding soils, consistent with brood cell depth (Potts and Willmer, 1997). Because surface and shallow soil temperatures are correlated (Rajver et al., 2006; West , 1952; Xu et al., 1997), they may provide reliable proxies of subsurface thermal conditions, potentially allowing females to use thermal cues when selecting nest sites. Exposure to sublethal desiccation stress did not alter CT_{max} (maximum critical temperature) or time to heat stupor in *L. malachurum* (Gonzalez et al., 2024a). Similarly, prior starvation followed by sublethal desiccation did not shift mean CT_{max}, but increased variance in CT_{max}, suggesting greater inter-individual variability in thermal tolerance (Gonzalez et al., 2024a). If similar variability is present among foundress females, it may influence which individuals successfully establish subsequent generations under acute thermal stress, with potential consequences for population dynamics.

4.5 Future directions

Personal observations during fieldwork and analysis of behavioural videos revealed that multiple individuals of the same species may share a nest entrance. Marking individuals would be needed to determine whether multiple foundresses contribute to nest establishment or merely share entrance access, as observed in other halictid bees (Paxton et al., 1999), and to test whether dominant females guard aggressively enough to exclude returning nestmates (Knerer , 1980; Knerer and Plateaux-Quénu , 1967; Lienhard et al., 2010). Velvet ants (Mutillidae), parasitic wasps (Hymenoptera), and bombyliid flies (Bombyliidae) (Batra , 1966; Lienhard et al., 2010) were frequently seen inspecting nesting sites, particularly in the afternoon when nests were often

closed; characterising brood-parasite presence and activity may therefore clarify nest-associated behaviour (Sick et al., 1994; Wcislo and Cane , 1996). Nest opening (Figure S6) and closing (Figure S7) durations and timings (Figure S8), and the conditions under which they occur, should also be recorded, as they may reveal the influence of temperature, predators, or nest-infesting parasites on nest closure and energy expenditure (Houston et al., 1993; Lienhard et al., 2010; Minchella , 1985; Sih and McCarthy , 2002; Vandame et al., 2002; Wuellner and Jang , 1996). Here, opening and closing could be observed only for *H. scabiosae*, whose nests were clearly visible: nest-opening duration varied from 2–118 s (mean ± SD: 32.2 ± 42.0 s; $n = 6$) and nest-closing duration from 15–456 s (170.0 ± 162.6 s; $n = 5$). Burrowing and nest-switching were not captured. Preliminary observations suggest that both species may inspect and enter multiple nests within an aggregation, particularly early in the nesting cycle, potentially facilitating nest switching or usurpation of unguarded nests, which appears more frequent in *H. scabiosae* (Knerer , 1969); mark–recapture approaches could quantify this, particularly given species differences in nest architecture (short laterals and sessile cells in *H. scabiosae*; brood cells at the ends of longer laterals in *L. malachurum*; (Knerer , 1969)).

Nesting habitat parameters such as soil type, texture, compaction, pH, aeration, bulk density, and moisture, which influence thermal diffusivity (Marshall et al., 1996), as well as microhabitat features including entrance orientation, vegetation cover, slope, shade, nearby stones (Antoine and Forrest , 2021; Cane , 2015; Jones and Oldroyd , 2006; Packer and Knerer , 1986b; Packer et al., 1989; Potts and Willmer, 1997; Shebl et al., 2016), and exposure to variable insolation (e.g. morning sun rather than total daily insolation; (Sakagami and Hayashida , 1961)), remain poorly understood in terms of their effects on bee fitness (Antoine and Forrest , 2021). Because nesting preferences cannot be generalised across ground-nesting bees (Harmon-Threatt , 2020), population-level studies are needed to assess inter-population variation. The initial days of nest founding in early spring, when weather is unpredictable and soils are colder than in summer, are poorly characterised, although soil temperature from nest founding to first brood in *L. malachurum* may influence colony development more than summer soil temperature (Weissel et al., 2006). Warmer soils may accelerate worker ageing and larval development and survival (Cossins , 2012; Jeanne and Morgan , 1992; Whitfield and Richards , 1992), shortening nesting cycles and pauses between broods (Weissel et al., 2006). A global review of 449 species of ground-nesting bees reported an average minimum nesting depth of approximately 17 cm (Cane and Neff , 2011), and colder soils at 20–40 cm slow nest construction in *L. malachurum* (Weissel et al., 2006); the 10 cm maximum depth sampled here may therefore not capture deeper nesting microclimates. Finally, because our two sites were only 7–10 m apart, comparisons across locations would help resolve site-specific effects (see Davis et al., 2019; Mikola et al., 2018; Redding et al., 2003; Stone , 1994).

4.6 Thermal limits and responses to climate change in bees

Temperature strongly influences the physiology, behaviour, mating (Larsson , 1991), energetic costs (Kearney et al., 2009), and life-history strategies such as timing of adult emergence (Forrest and Chisholm , 2017), number of generations per year (Forrest et al., 2019), and larval maturation and development in insects (Borrell and Medeiros, 2004; Grafton-Cardwell et al., 2005; Woods

et al., 2005). Warmer temperatures can lead to shorter nesting cycles and a higher number of generations per year, but may also result in smaller body size and increased mortality in ectotherms; thus, population dynamics and fitness consequences are highly complex and context-dependent (Forrest and Chisholm , 2017; Kingsolver and Huey , 2008). Alongside rising temperatures, a suite of anthropogenic stressors—including habitat loss, pesticide exposure, and environmental disturbances such as wildfires, whose frequency is increasing, shifts in vegetation cover driven by elevated atmospheric CO_2 (Hoffmann et al., 2002), flooding events (Potts et al., 2016; Roubik and Michener , 1980), reduced floral resources due to invasive species, and pathogen and parasite spillover from managed bees—represent major contemporary threats to bees. Maintaining and restoring native vegetation can provide thermal refuges during heat stress or heat waves. At our site, which showed patchy anthropogenic disturbance and marked day-to-day variation in temperature and rainfall (Supplementary Table S1), conditions for ground-nesting bees were highly variable; coping with and adapting life-history strategies to such conditions is essential for survival and reproduction. During the study period, rain fell on three days (maximum 10.6 mm in a single day; total 17.4 mm; Supplementary Table S1), within the highly variable rainfall regime characteristic of the Mediterranean climate of Lesvos. *H. scabiosae* has recently been reported to expand its range into previously unsuitable habitats in response to environmental change, with urbanisation playing a major role in this process; however, its ecological niche—characterised by high temperature seasonality and low precipitation—has remained relatively stable from 1970 to the present, suggesting limited adaptability to novel conditions (Gil-Tapetado et al. , 2024). Consequently, habitat fragmentation and increasing urban density (Lanner et al. , 2022; Polidori et al. , 2021) may facilitate colonisation of previously unoccupied areas by *H. scabiosae* (Gil-Tapetado et al. , 2024), although not all bee species, particularly those with specialised resource use and limited dispersal capacity, can track suitable climates by shifting poleward or to higher latitudes (Buckner and Danforth , 2022; Casey et al. , 2015; Kerr et al. , 2015; Martínez-López et al. , 2021; Nemésio et al. , 2016). The progressively earlier onset of spring in recent years highlights how shifting environmental conditions may require bees to adjust the timing of seasonal activity and termination (Kearney et al., 2009) and adopt novel behavioural strategies to cope with thermal and climatic variability.

Over evolutionary timescales, species have shifted their geographic distributions, and such shifts are expected to become more frequent under ongoing climate change. However, physiological heat tolerance is often conserved across lineages and constrained by relatively fixed upper limits (Araújo et al., 2013; Hoffmann et al., 2013). The CT_{max} values of *H. scabiosae* (43 °C) and *L. malachurum* (42 °C) (Gonzalez et al., 2024) exceed mean summer maximum air temperatures on Lesvos (28.2 °C, June–August 2003–2019; Supplementary Table S3), but mean maxima do not capture heat waves or microhabitat extremes, which may bring individuals closer to their limits (Clusella-Trullas et al., 2011; Deutsch et al., 2008; Huey et al., 2009). Behavioural thermoregulation can buffer individuals against exceeding physiological limits and increase short-term persistence, but it is ultimately constrained by those limits (Huey et al., 2003), which in turn restrict how far realised niches can expand beyond conserved fundamental thermal boundaries (Jackson and Overpeck , 2000; Kearney et al., 2013). Dispersal limitation (Baselga et al., 2012), historical contingency (Ricklefs , 2006), and biotic interactions (Dobzhansky , 1950) fur-

ther shape realised distributions (Araújo et al., 2013). Because our data concern recovery from cold in spring rather than heat tolerance, they speak most directly to how early-season thermal conditions shape activity; whether the species differences we observed extend to upper thermal limits remains to be tested.

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Table S1: Daily climatological summary for 15–24 April 2026 (Molyvos, Lesvos; 290 m above sea level). Mean air temperature (T_{mean}) over the period was 14.63 °C, and mean relative humidity (RH; calculated as the mean of daily maximum and minimum values) was 67.9%. Total rainfall was 17.4 mm, recorded on 15, 22, and 23 April; maximum daily rainfall was 10.6 mm (22 April). Data source: Meteo Aegean meteorological database (`http://meteo.aegean.gr`).

Day	T_{mean} (°C)	T_{max} (°C)	Time of T_{max}	T_{min} (°C)	Time of T_{min}	$\mathrm{RH}_{\mathrm{max}}$ (%)	$\mathrm{RH}_{\mathrm{min}}$ (%)	Rain (mm)
15	16.3	19.8	15:00	13.8	02:20	79	60	0.4
16	17.8	23.7	15:20	13.5	23:50	75	43	0.0
17	13.1	17.2	14:30	10.3	07:20	85	63	0.0
18	14.1	19.9	14:50	10.3	06:10	79	36	0.0
19	13.7	19.4	13:00	8.6	06:20	80	47	0.0
20	15.9	21.1	12:00	9.1	05:10	75	45	0.0
21	17.3	21.1	15:50	13.3	05:50	84	53	0.0
22	14.3	18.1	15:50	10.2	23:40	93	71	10.6
23	11.2	16.0	17:30	8.1	00:00	93	60	6.4
24	12.6	17.2	12:20	6.9	01:20	82	55	0.0

Table S2: Multi-year April climatological summary for the 15 years with data available between 2003 and 2019 (2003 and 2007 omitted), from the Agia Paraskevi meteorological station, the closest available long-term weather station to Skala Kallonis on Lesvos. Values are April monthly means for each year. Across the years shown, mean air temperature (T_{air}) was 14.6 °C, mean relative humidity (RH) was 65.9%, and mean soil temperature (T_{soil}) was 14.4 °C. Data source: Meteo Aegean meteorological database (`http://meteo.aegean.gr`).

Year	T_{air} (°C)	RH (%)	T_{soil} (°C)
2019	14.4	75.4	14.0
2018	18.3	62.8	14.0
2017	15.1	64.2	14.9
2016	17.6	64.4	16.2
2015	13.3	68.5	12.1
2014	15.7	73.2	15.0
2013	16.0	64.8	15.5
2012	14.5	73.0	15.1
2011	11.4	73.9	12.7
2010	14.3	61.8	14.4
2009	13.5	61.9	13.2
2008	14.2	69.8	14.2
2006	14.3	55.2	15.8
2005	13.0	53.4	14.5
2004	13.1	65.6	14.7

Table S3: Annual summer maximum air temperatures at the Agia Paraskevi meteorological station, the closest long-term climate station to Skala Kallonis on Lesvos. For each year, the table shows the highest monthly mean daily maximum air temperature in June–August and the month in which it occurred. Mean annual summer maximum temperature across 2003–2019 was 28.2 °C. Data source: Meteo Aegean meteorological database (`http://meteo.aegean.gr`).

Year	Month of highest value	Highest monthly mean of daily maximum air temperature (°C)
2003	August	27.4
2004	July	26.0
2005	July	27.0
2006	July	25.7
2007	June	28.6
2008	August	27.4
2009	July	27.0
2010	August	29.0
2011	July	27.0
2012	July	32.1
2013	August	29.0
2014	August	29.0
2015	August	29.4
2016	July and August	29.0
2017	July	28.3
2018	August	29.0
2019	August	29.0

Table S4: Summary of focal-view behavioural metrics for HSC (*H. scabiosae*) and LMA (*L. malachurum*). Activity is the mean number of bees visible outside nests per observation interval; guarding time is the mean duration in seconds. Nest-entry and nest-exit frequencies are mean counts per 10-min focal-view recording. Nest guarding was not assessed for LMA.

Species	Activity	Guarding time (s)	Nest-entry frequency	Nest-exit frequency
HSC	0.969	186.557	0.626	0.491
LMA	0.254	–	0.307	0.245

Table S5: Robustness checks for pooled wide-angle generalised additive models (GAMs), fitted to 166 recordings. Within-day variation is the percentage of temperature variation occurring within sampling days. The AIC reduction compares models with and without a sampling-date random intercept; positive values indicate lower AIC when date is included. AIC, Akaike's information criterion.

Predictor	Temperature variation within days (%)	AIC reduction with date	Smooth-term
Ambient air temperature	59.2	37.50	
Shallow soil temperature (2–3 cm)	72.2	54.20	
Middle soil temperature (5 cm)	75.5	48.46	
Deep soil temperature (10 cm)	96.3	93.39	

Table S6: DHARMa residual diagnostics for focal-view entry and exit, and wide-angle generalised additive models (GAMs). Entries are p-values for tests of residual uniformity, dispersion, and outliers.

Model	Uniformity (Kolmogorov–Smirnov test)	Dispersion	Outliers
Focal: *H. scabiosae* nest entry	0.018	0.520	1.000
Focal: *H. scabiosae* nest exit	0.034	0.604	1.000
Focal: *L. malachurum* nest entry	< 0.001	0.536	0.016
Focal: *L. malachurum* nest exit	< 0.001	0.540	0.016
Pooled: ambient air temperature	0.995	0.528	1.000
Pooled: shallow soil temperature	0.842	0.482	1.000
Pooled: middle soil temperature	0.412	0.482	1.000
Pooled: deep soil temperature	0.666	0.454	1.000
Site-specific: ambient air temperature	0.785	0.584	1.000
Site-specific: shallow soil temperature	0.191	0.594	1.000
Site-specific: middle soil temperature	0.640	0.390	1.000
Site-specific: deep soil temperature	0.836	0.346	1.000
Within-day: ambient air temperature	0.982	0.520	1.000
Within-day: shallow soil temperature	0.531	0.556	1.000
Within-day: middle soil temperature	0.377	0.508	1.000
Within-day: deep soil temperature	0.219	0.482	1.000

Table S7: Species means for chill-coma recovery and take-off traits in HSC (*H. scabiosae*; $n = 18$) and LMA (*L. malachurum*; $n = 17$). ITD, intertegular distance; T_R, righting temperature; D_R, time to right; T_T, take-off temperature; D_T, time to take-off.

Species	ITD (cm)	T_R (°C)	D_R (s)	T_T (°C)	D_T (s)
HSC	0.340	29.8	185	32.8	505
LMA	0.206	26.4	131	31.2	468

Table S8: Linear mixed-effects models (LMMs) examining physiological and behavioural performance traits in HSC (*H. scabiosae*) and LMA (*L. malachurum*). Sampling date was included as a random intercept in all models, and HSC was the reference level for species comparisons. Intertegular distance (ITD) was excluded from these models because it was collinear with species identity. Results are reported as regression coefficients (β), standard errors, t-statistics, and p-values. Random-intercept variance for sampling date was 1.148 for righting temperature, 0 for take-off temperature, 0.066 for time to right, and 0 for time to take-off. The take-off temperature and time-to-take-off models had singular fits, indicating negligible estimated variance among sampling dates. The exploratory rows come from separate linear models for each species (n = 18 HSC, 17 LMA) that also included ambient temperature (coefficients not shown); ITD coefficients are per cm. ITD was not associated with time to right or time to take-off in either species (all $p \geq 0.24$). Sensitivity models that also included log-transformed time to right or time to take-off are reported in the text.

Response variable	Predictor	β	Standard error	Test s
Righting temperature	LMA (vs HSC)	-2.488	0.876	t =
	Ambient temperature	0.977	0.149	t
Take-off temperature	LMA (vs HSC)	-2.100	1.115	t =
	Ambient temperature	1.413	0.234	t
Time to right (log-transformed)	LMA (vs HSC)	-0.422	0.214	t =
	Ambient temperature	0.033	0.036	t
Time to take-off (log-transformed)	LMA (vs HSC)	0.003	0.251	t
	Ambient temperature	0.058	0.053	t
Righting temperature, within species (exploratory)	ITD, HSC only	14.26	12.14	t
	ITD, LMA only	-93.07	36.45	t =
Take-off temperature, within species (exploratory)	ITD, HSC only	-9.79	22.69	t =
	ITD, LMA only	-91.60	41.59	t =

Table S9: Daily mean number of bees observed outside nests per observation interval in the wide-angle recordings from 15–24 April 2026. Values show mean activity, standard deviation, and number of recordings for each date.

Date	Mean activity	Standard deviation	Number of recordings
15 Apr	1.36	0.91	16
16 Apr	0.78	0.82	16
17 Apr	0.84	0.67	16
18 Apr	1.49	1.16	16
19 Apr	1.40	1.06	16
20 Apr	1.55	1.41	15
21 Apr	1.10	1.12	18
22 Apr	0.47	0.51	17
23 Apr	0.17	0.26	18
24 Apr	0.99	1.07	18

Table S10: Pooled generalised additive models (GAMs) of bee activity in relation to each temperature predictor, fitted to 166 wide-angle recordings. Each model included site as a fixed effect and sampling date as a random intercept. EDF, effective degrees of freedom.

Predictor	EDF	F	p	Adjusted R^2	Deviance explained (%)
Ambient air temperature	2.95	22.63	< 0.001	0.427	46.7
Shallow soil temperature (2–3 cm)	3.48	11.72	< 0.001	0.342	39.3
Middle soil temperature (5 cm)	3.64	13.26	< 0.001	0.357	40.5
Deep soil temperature (10 cm)	3.83	38.11	< 0.001	0.569	60.4

Table S11: Linear models (LMs), linear mixed-effects models (LMMs), and generalised linear mixed-effects models (GLMMs) linking environmental predictors to nest activity and abundance in *H. scabiosae* (HSC) and *L. malachurum* (LMA). Sampling day and/or site were included as random effects in mixed-effects models. GLMM results are reported as log-odds with Wald z-statistics; LM and LMM results are reported as regression coefficients with t-statistics. Environmental predictors in the species-combined abundance model were standardised. Their main-effect coefficients represent slopes for HSC, and interaction coefficients represent differences in slope for LMA relative to HSC.

Response variable	Model	Predictor	β	Standard error	Test statistic	p
Focal-view occurrence models						
Guard presence (HSC)	GLMM	Shallow temperature (scaled)	-0.942	0.269	$z = -3.50$	< 0.001
Nest guarding duration (HSC, log)	LMM	Middle temperature (scaled)	0.506	0.166	$t = 3.04$	0.004
Abundance model (species-combined)						
$\log(\text{abundance} + 1)$	LM	Species (LMA vs HSC)	-0.368	0.051	$t = -7.25$	< 0.001
		Ambient temperature	-0.028	0.056	$t = -0.51$	0.613
		Shallow temperature	0.318	0.083	$t = 3.85$	< 0.001
		Middle temperature	-0.289	0.103	$t = -2.81$	0.006
		Deep temperature	-0.045	0.077	$t = -0.59$	0.558
		Light intensity	0.242	0.051	$t = 4.74$	< 0.001
		Species × ambient temperature	0.088	0.079	$t = 1.11$	0.269
		Species × shallow soil temperature	-0.223	0.117	$t = -1.91$	0.058
		Species × middle soil temperature	0.169	0.145	$t = 1.16$	0.248
		Species × deep soil temperature	0.088	0.109	$t = 0.81$	0.421
		Species × light intensity	-0.207	0.072	$t = -2.87$	0.005

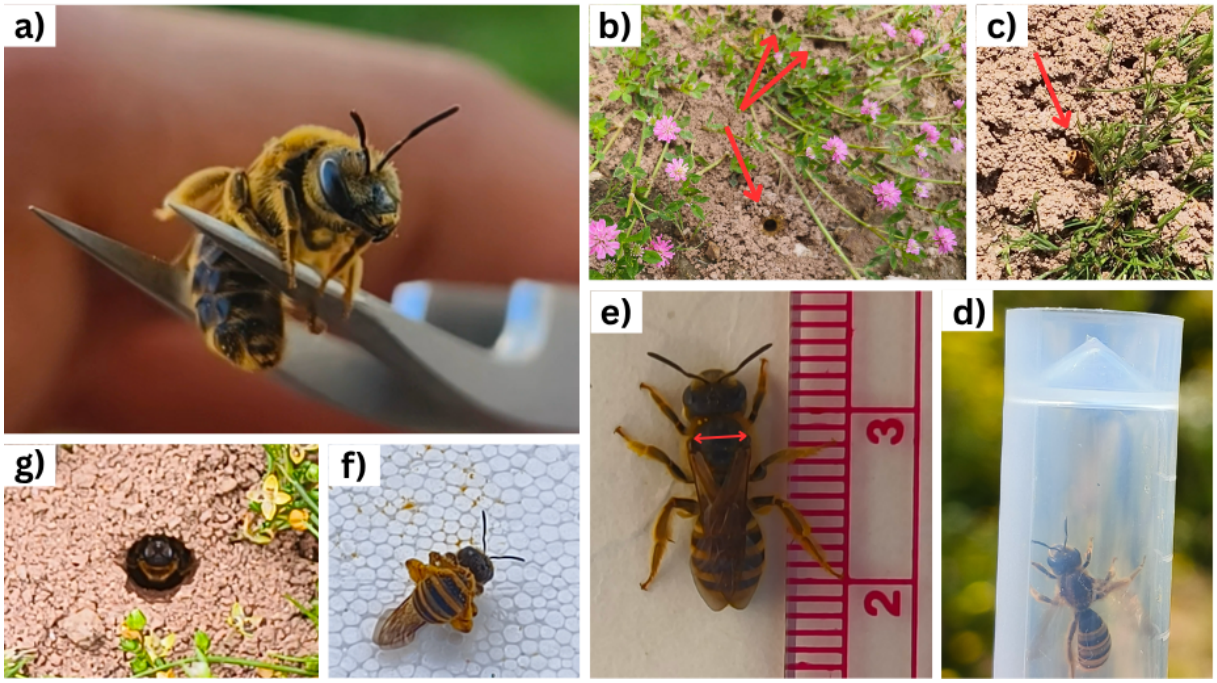


Figure S1: Field observations and chill-coma recovery assays in *H. scabiosae*. (a) Adult in profile. (b) Nesting aggregation among Persian clover (*Trifolium resupinatum*); black lines mark open nest entrances. (c) An individual entering a nest, marked by a black line. (d) Capture with an insect net and transfer to a plastic vial for the recovery assay. (e) An immobilised individual positioned dorsal-side up beside a scale for morphometric analysis in ImageJ; the red line indicates intertegular distance (ITD). (f) An individual recovering in a ventral-side-up posture on an insulated surface; the curled abdomen is characteristic of recovery behaviour in this species. (g) An individual guarding a nest entrance.

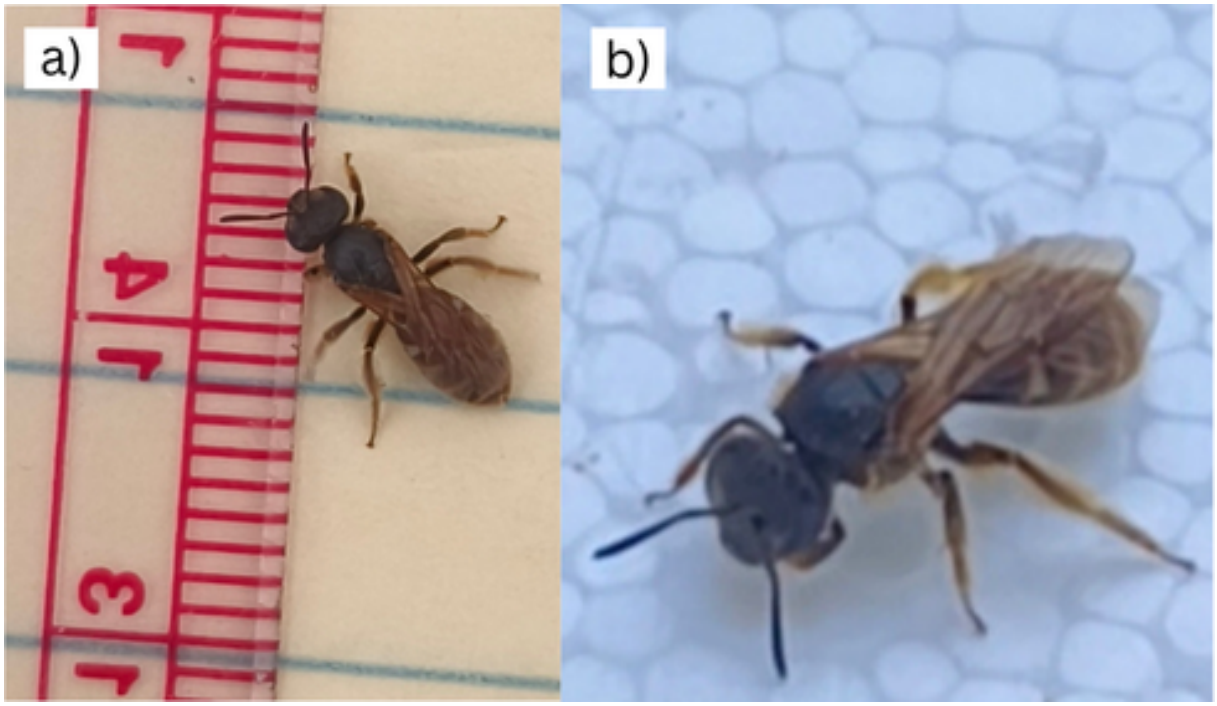


Figure S2: *L. malachurum* during chill-coma recovery assays. (a) An individual positioned ventral-side down beside a scale for image-based measurement of intertegular distance. (b) An individual that has righted itself but has not yet initiated flight.

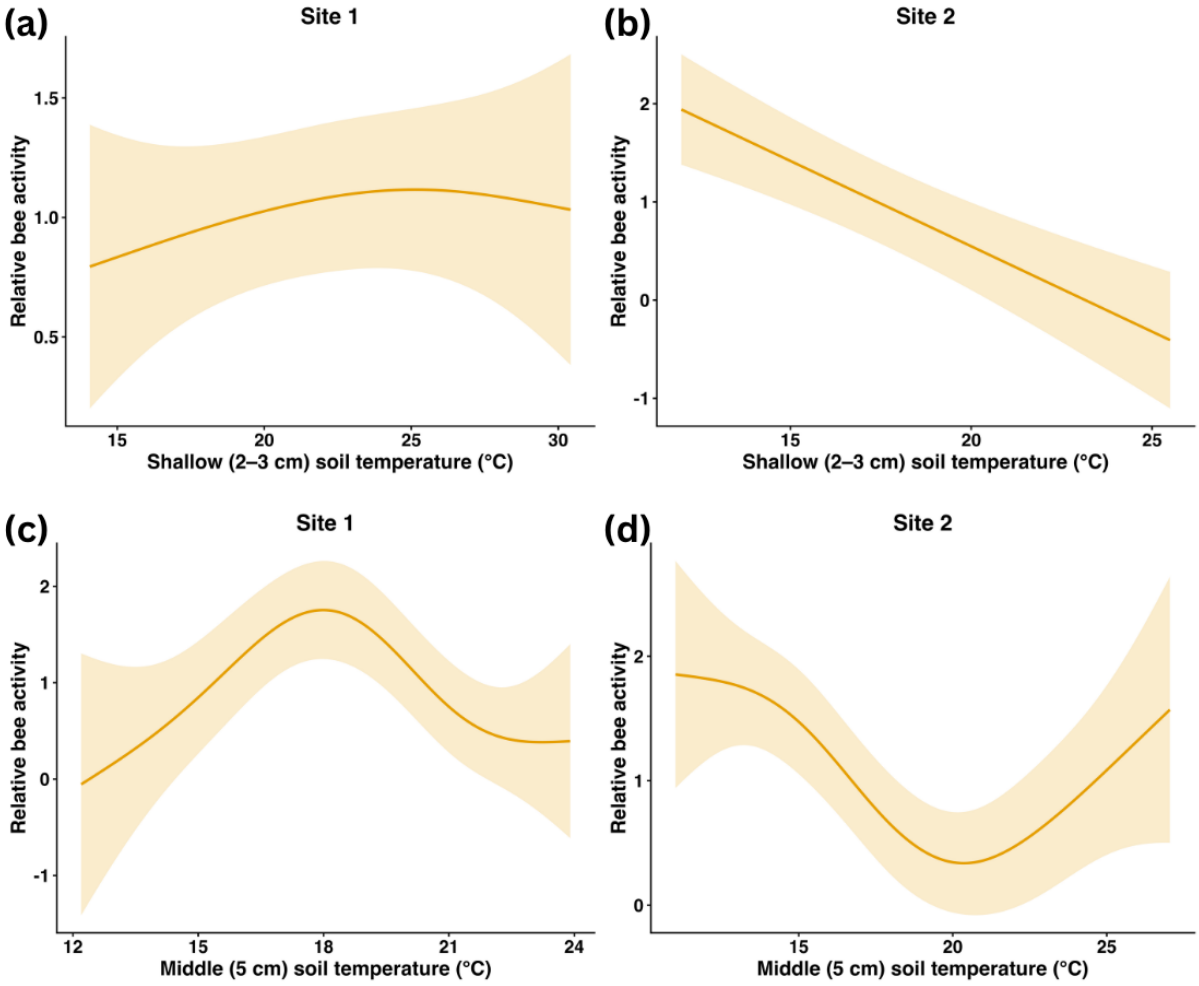


Figure S3: Site-specific generalised additive model (GAM) smooths showing the relationship between wide-angle bee activity (mean number of bees observed) and shallow soil temperature (2–3 cm) and middle soil temperature (5 cm) at Sites 1 and 2. Shaded ribbons indicate 95% confidence intervals.

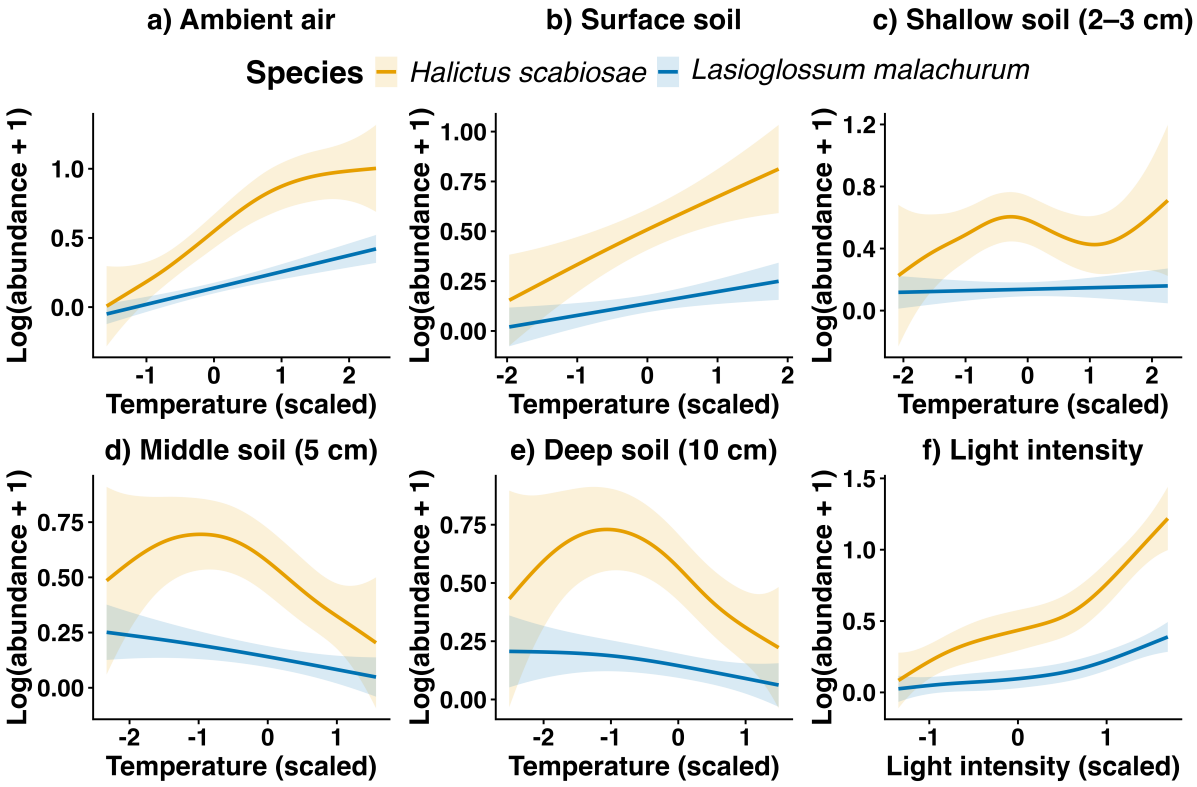


Figure S4: Species-specific generalised additive model (GAM) smooths relating abundance (log(abundance+1)) of *H. scabiosae* and *L. malachurum* to (a) ambient air temperature, (b) surface soil temperature, (c) shallow soil temperature (2–3 cm), (d) middle soil temperature (5 cm), (e) deep soil temperature (10 cm), and (f) light intensity (all scaled). Each environmental variable was fitted in a separate single-predictor model for each species, using the 95 focal-view observations with complete logger data. Shaded ribbons indicate 95% confidence intervals.

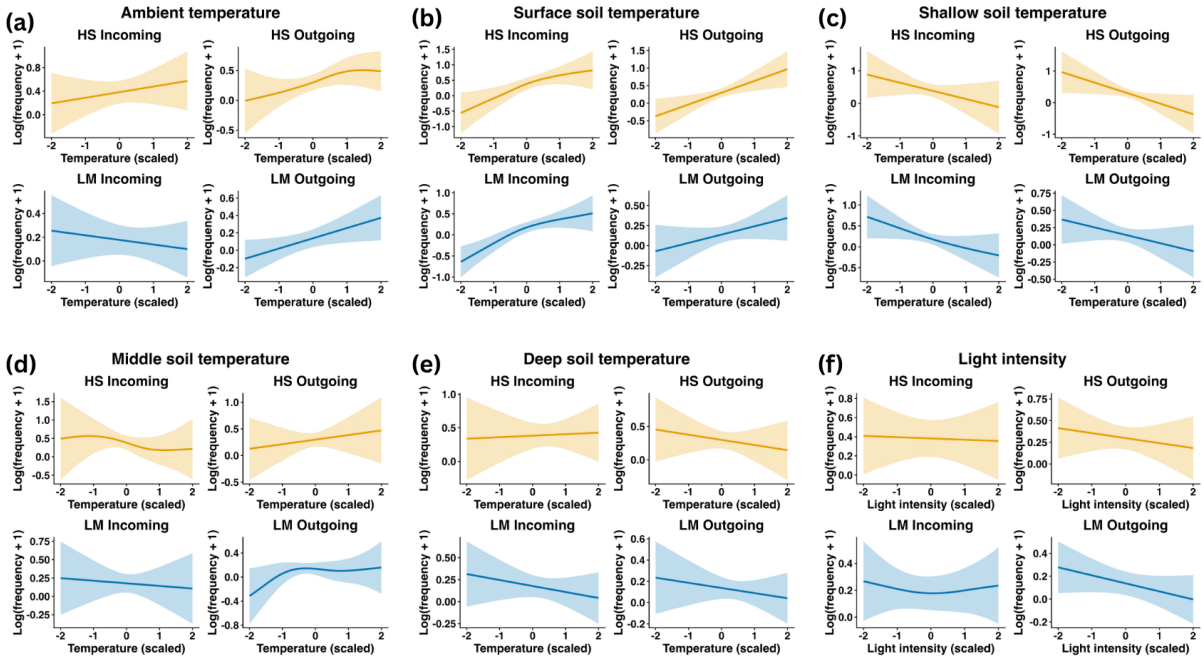


Figure S5: Species-specific generalised additive model (GAM) smooths relating nest entry frequency (entries per 10-min focal-view recording) in *H. scabiosae* and *L. malachurum* to (a) ambient air temperature, (b) surface soil temperature, (c) shallow soil temperature (2–3 cm), (d) middle soil temperature (5 cm), (e) deep soil temperature (10 cm), and (f) light intensity (all scaled). Each environmental variable was fitted in a separate single-predictor model for each species, using the 95 focal-view observations with complete logger data. Shaded ribbons indicate 95% confidence intervals.

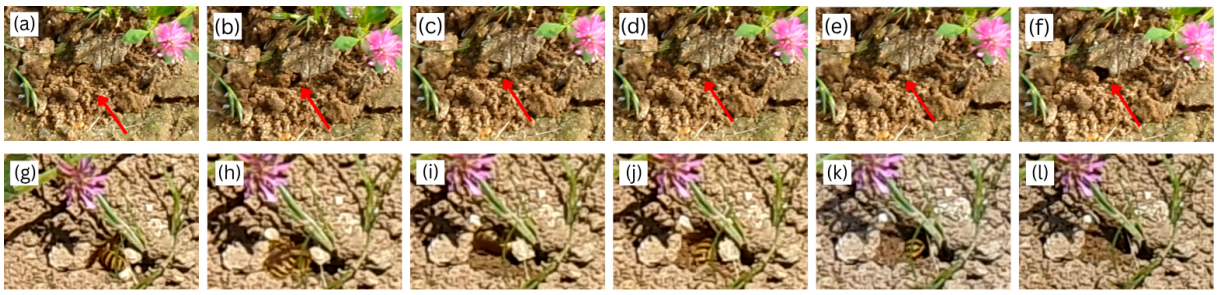


Figure S6: Nest opening and new nest excavation behaviour in *H. scabiosae* at Site 2. (a–f) Nest opening recorded on 15 April during the 10:00–11:00 h observation period. Timestamps are minutes:seconds after 10:00 h: (a) 00:00, (b) 08:03, (c) 09:46, (d) 09:52, (e) 11:56, and (f) 12:47; the camera continued recording after the 10-min analysed window. The sequence from (b) to (f) spans approximately 4 min 44 s. Red arrows mark the nest entrance; an individual at the entrance is visible in panels (c)–(e). (g–l) Excavation of a new nest entrance in loose soil recorded on 19 April during the 13:00–14:00 h observation period. Timestamps are minutes:seconds after 13:00 h: (g) 00:41, (h) 01:19, (i) 02:37, (j) 03:22, (k) 04:36, and (l) 04:47. The sequence spans approximately 4 min 06 s; the excavating individual is visible in panels (g)–(k).

Table S12: Generalised additive models (GAMs) of nest-entry and nest-exit frequency in *H. scabiosae* (HSC) and *L. malachurum* (LMA), fitted to 95 focal-view observations. Each model included smooth terms for all six environmental predictors. Frequencies are counts of nest entries or exits per 10-min focal-view recording. EDF, effective degrees of freedom.

Response	Predictor	EDF	Reference degrees of freedom	F	p	Deviance explained (%)
HSC nest-entry frequency	Ambient air temperature	1.00	1.00	0.63	0.429	40.5
	Surface soil temperature	2.13	2.74	3.54	0.024	40.5
	Shallow soil temperature	1.00	1.00	1.76	0.188	40.5
	Middle soil temperature	2.54	3.28	1.25	0.246	40.5
	Deep soil temperature	1.00	1.00	0.03	0.859	40.5
	Light intensity	1.00	1.00	0.02	0.883	40.5
HSC nest-exit frequency	Ambient air temperature	2.24	2.83	1.00	0.333	28.7
	Surface soil temperature	1.00	1.00	7.43	0.008	28.7
	Shallow soil temperature	1.00	1.00	4.46	0.038	28.7
	Middle soil temperature	1.00	1.00	0.33	0.568	28.7
	Deep soil temperature	1.00	1.00	0.47	0.497	28.7
	Light intensity	1.00	1.00	0.46	0.501	28.7
LMA nest-entry frequency	Ambient air temperature	1.00	1.00	0.40	0.527	34.1
	Surface soil temperature	2.36	3.00	6.32	< 0.001	34.1
	Shallow soil temperature	1.58	1.95	2.23	0.143	34.1
	Middle soil temperature	1.00	1.00	0.09	0.769	34.1
	Deep soil temperature	1.00	1.00	0.75	0.390	34.1
	Light intensity	1.48	1.79	0.31	0.725	34.1
LMA nest-exit frequency	Ambient air temperature	1.00	1.00	4.61	0.035	23.0
	Surface soil temperature	1.00	1.00	1.98	0.163	23.0
	Shallow soil temperature	1.00	1.00	1.64	0.203	23.0
	Middle soil temperature	3.16	4.01	2.44	0.052	23.0
	Deep soil temperature	1.00	1.00	0.47	0.494	23.0
	Light intensity	1.00	1.00	2.01	0.160	23.0

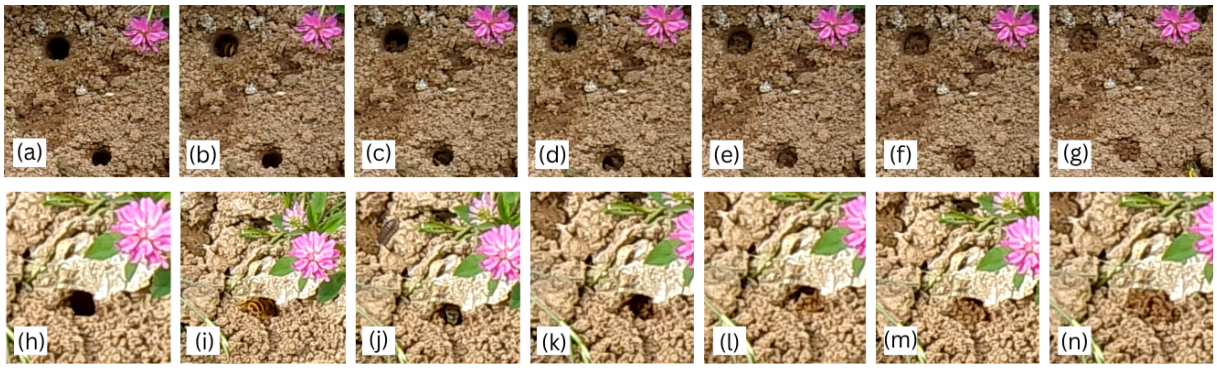


Figure S7: Nest-closing behaviour in *H. scabiosae* at Site 2. (a–g) Simultaneous closure of two nest entrances on 17 April during the 13:00–14:00 h observation period. Times are elapsed from the start of the recording: (a) 07:31, (b) 07:49, (c) 08:17, (d) 08:26, (e) 08:28, (f) 08:50, and (g) 09:23. The sequence spans approximately 1 min 52 s. (h–n) Nest-closing behaviour on 15 April during the 16:00–17:00 h observation period. Times are elapsed from the start of the recording: (h) 00:36, (i) 01:21, (j) 02:20, (k) 04:35, (l) 05:18, (m) 06:33, and (n) 09:26. The sequence spans approximately 8 min 50 s.



Figure S8: Hourly changes in nest entrance status within the mixed-species aggregation at Site 1. Panels show the same field of view at (a) 10:00 h, (b) 11:00 h, (c) 12:00 h, (d) 13:00 h, and (e) 14:00 h. Red arrows mark open or partially open entrances of *H. scabiosae* and *L. malachurum*.